Analyse des données tests 1 à 14 réalisés le 10 avril 2025, en présence de la structure

Calculs pour chaque test reportés ci après:

- Amplitude de l'onde incidente en amont
- Coefficient de réflexion en amont
- Amplitude de l'onde transmise
- Réflexion de la plage.

A partir des amplitudes et phases mesurées à la période de l'onde à partir du calcul des coefficients de Fourier.

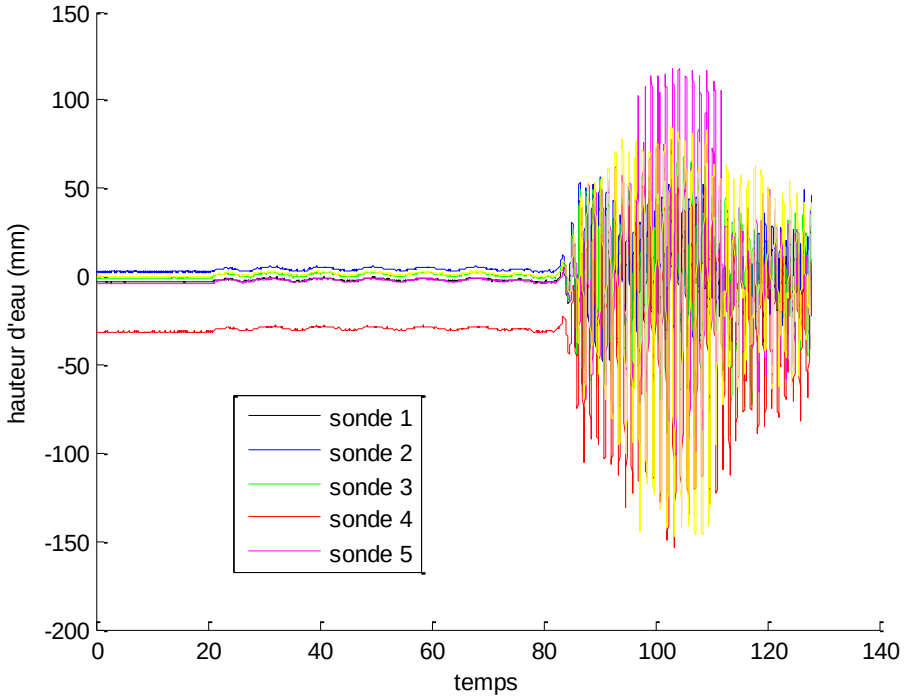
Utilisation pour la séparation des ondes incidente et réfléchie:
En amont, sondes 2, 3 et 4
En aval, sondes 11, 13 et 14

Affichage des signaux amont et aval, puis des signaux pour la 3eme sonde, mesurés et calculés par la méthode de Fourier.
L'affichage des signaux pour la 3eme sonde permet de voir la fenêtre de la série temporelle utilisée pour les calculs

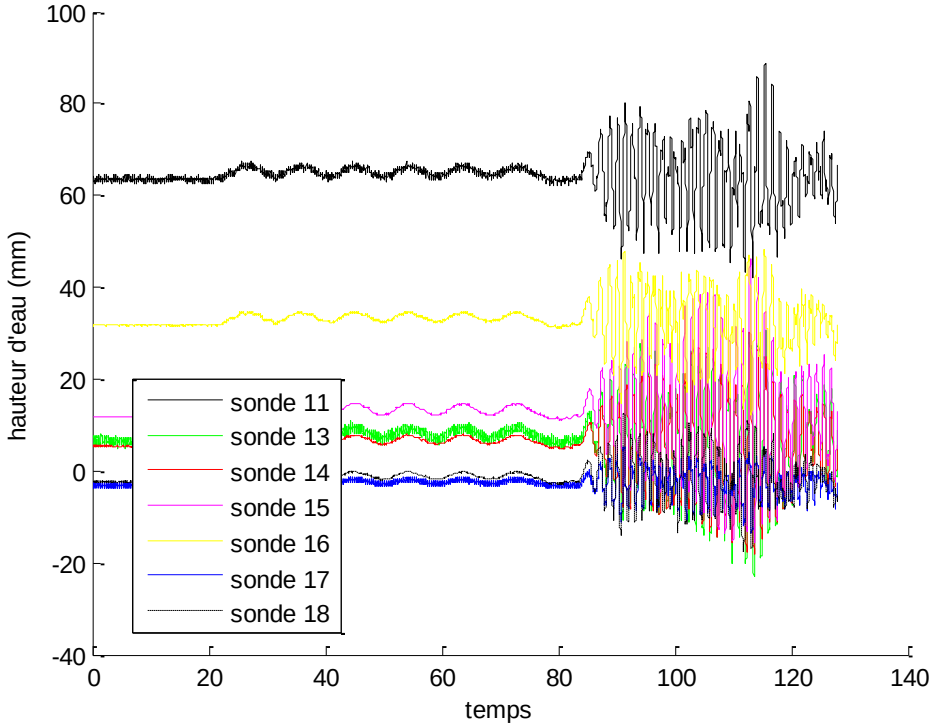
test_1_Hs_2_1_Tp_5_5

Tm=1.23s

Amont
Signal des sondes



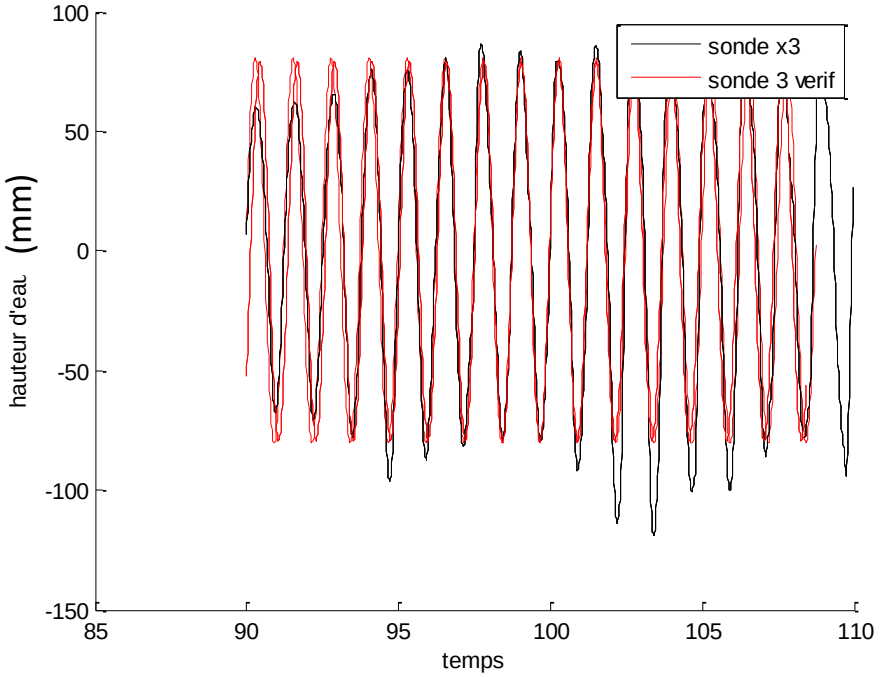
Aval
Signal des sondes



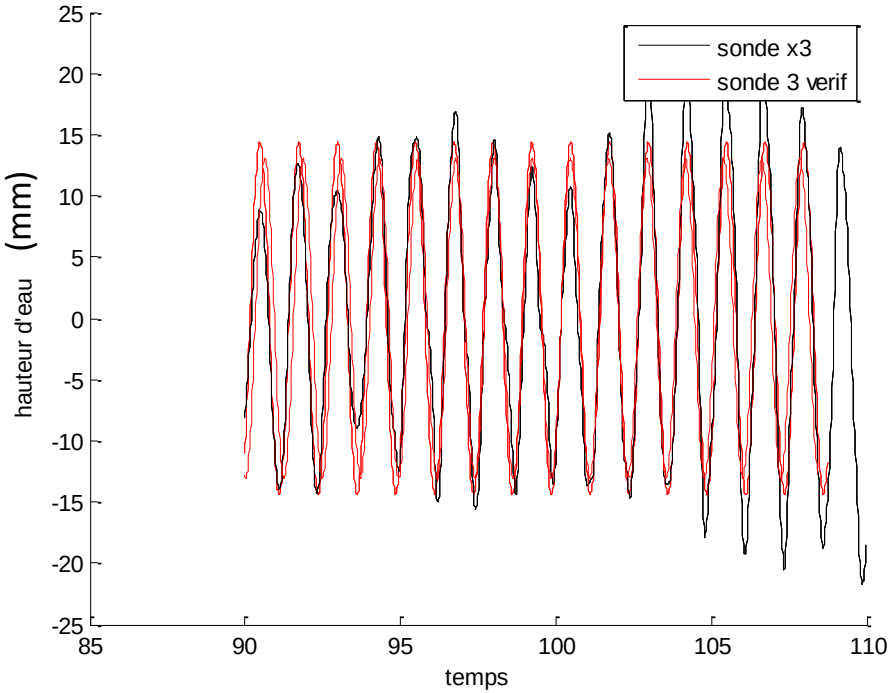
test_1_Hs_2_1_Tp_5_5

Tm=1.23s

amont



aval



aincident = 59.9803 mm

refl_3sondes_structure = 0.4169

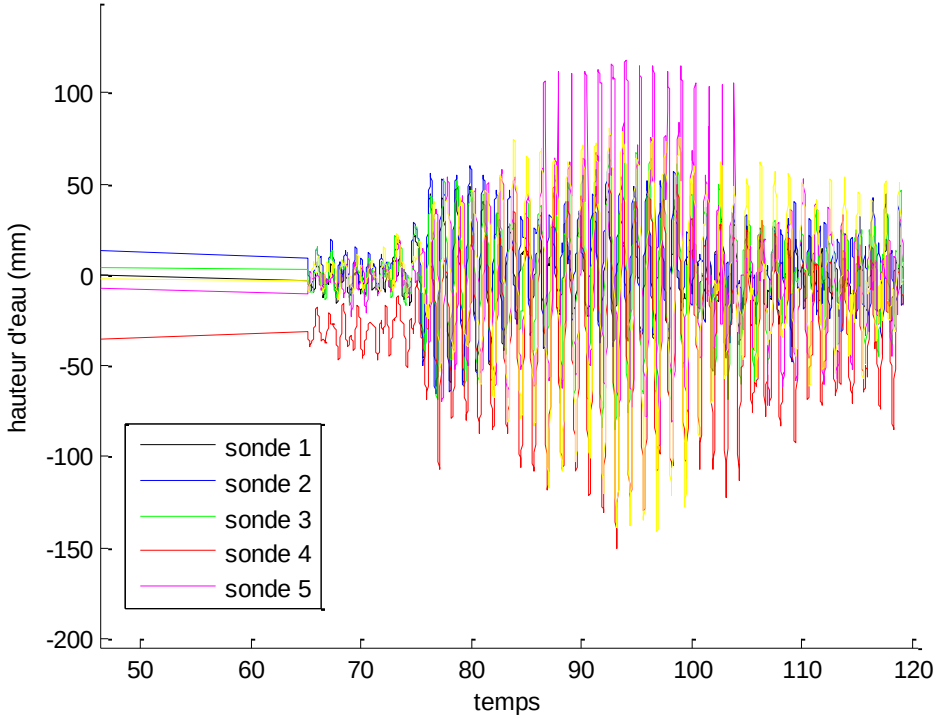
atransmis= 11.3772 mm

refl_s3sondes_plage = 0.1817

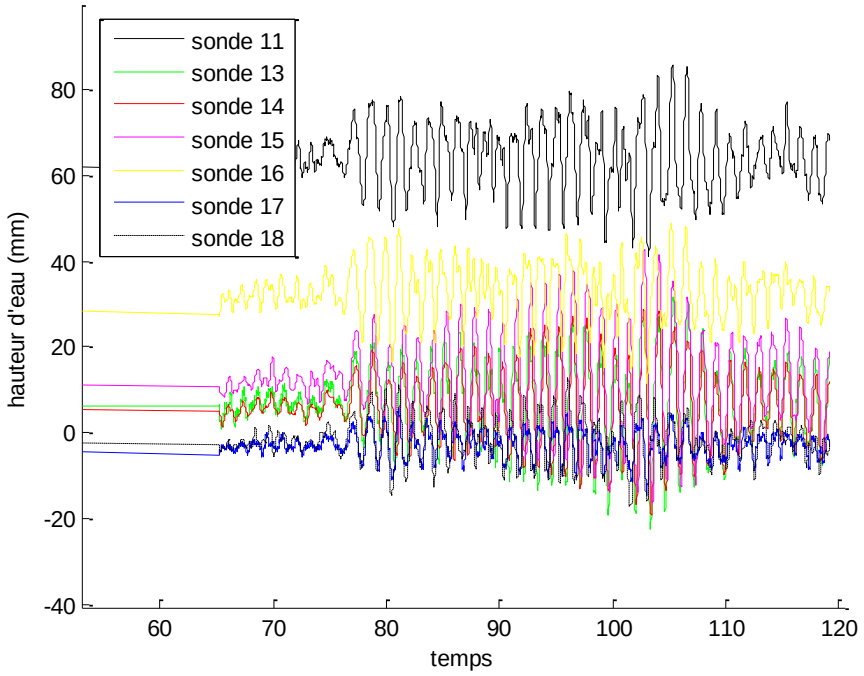
test_2_Hs_2_0_Tp_5_5

Tm=1.23s

Amont
Signal des sondes



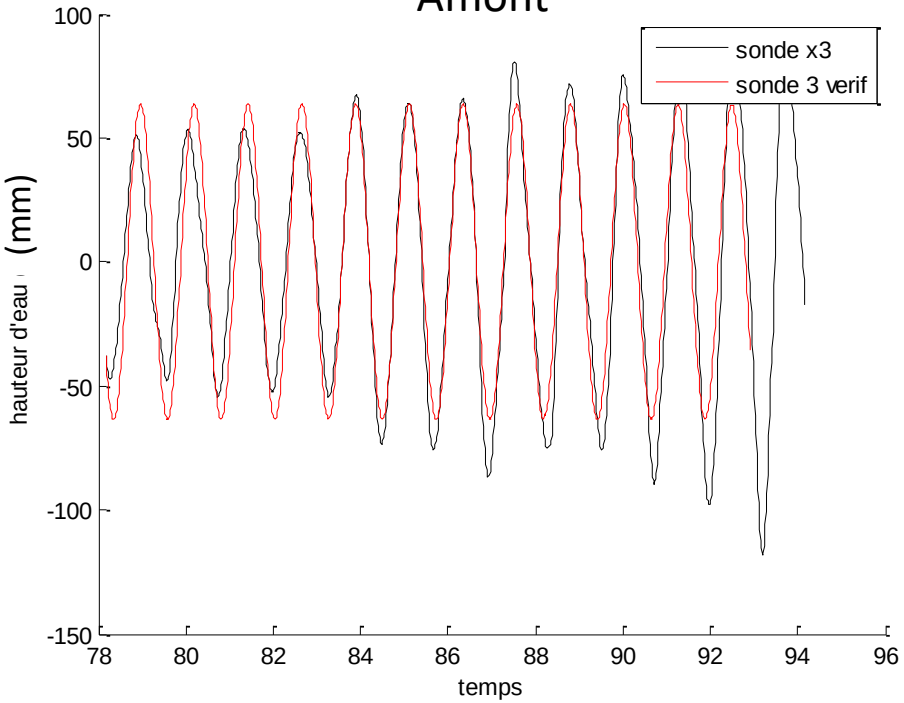
Aval
Signal des sondes



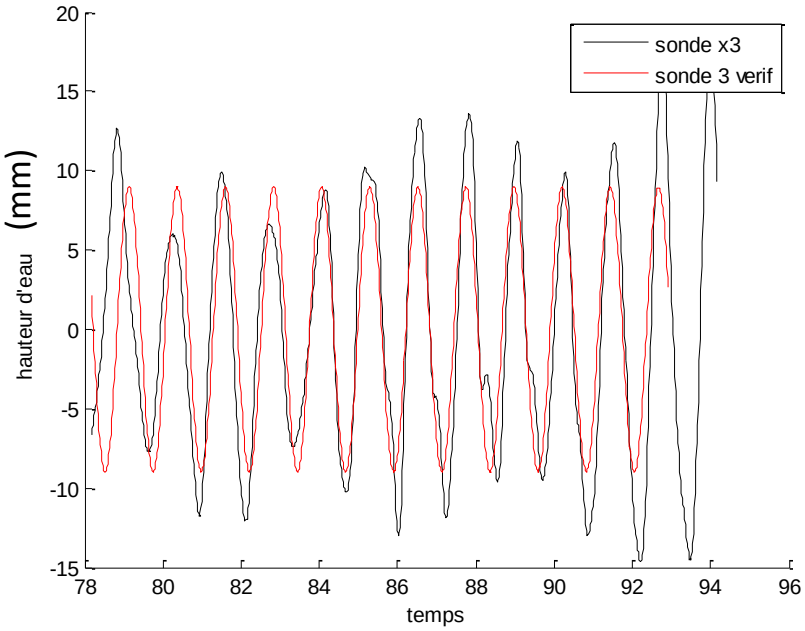
test_2_Hs_2_0_Tp_5_5

Tm=1.23s

Amont



Aval



aincident =52.1627mm

a_transmis= 8.6238mm

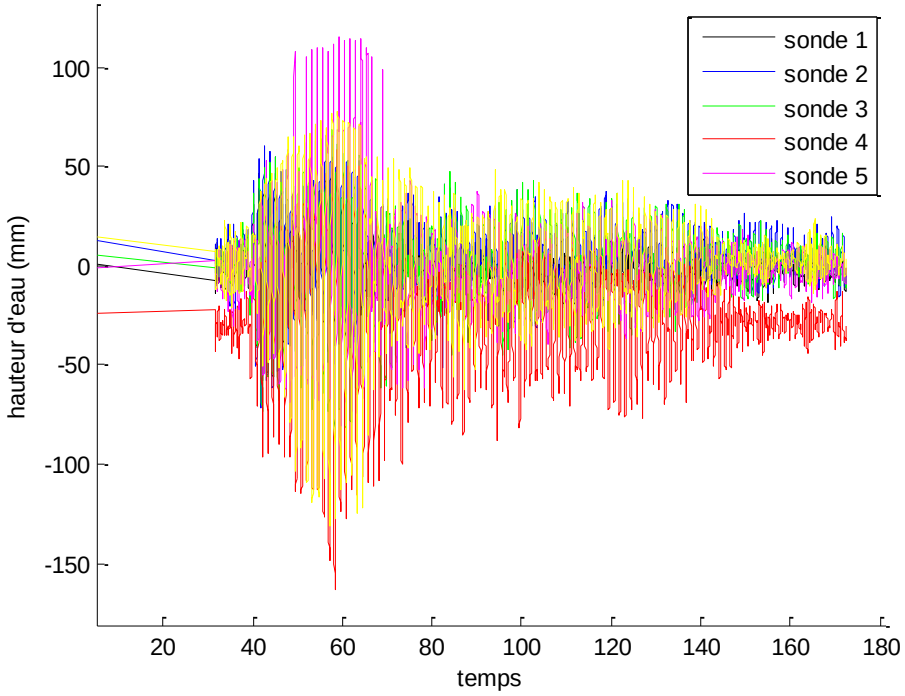
refl_s3sondes = 0.3217

refl_s3sondes_plage =0.0875

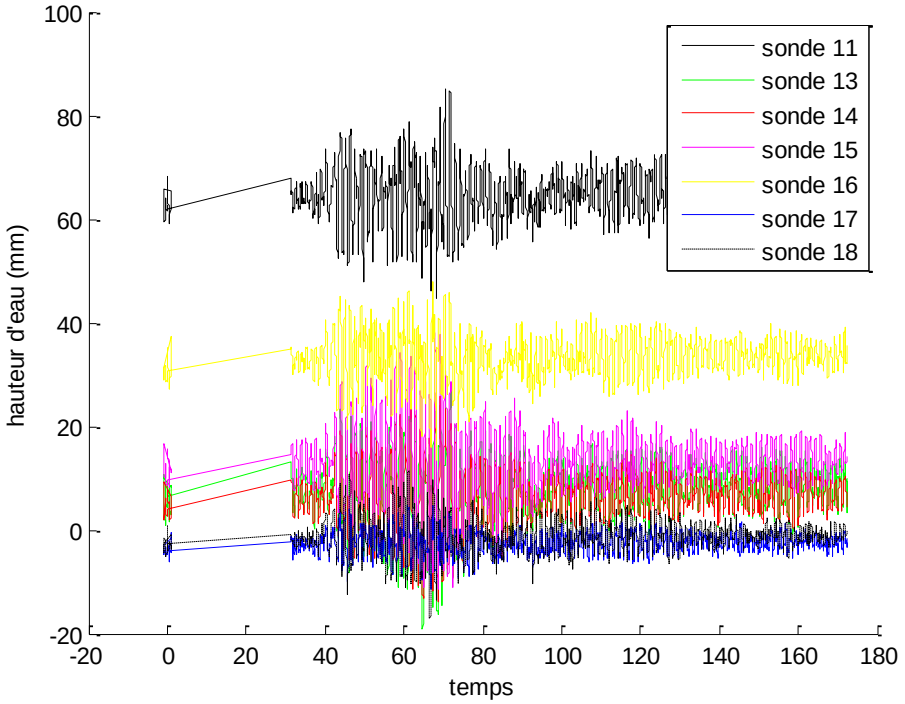
test_3_HS_1_8_Tp_5_5

Tm=1.23s

Amont
Signal des sondes



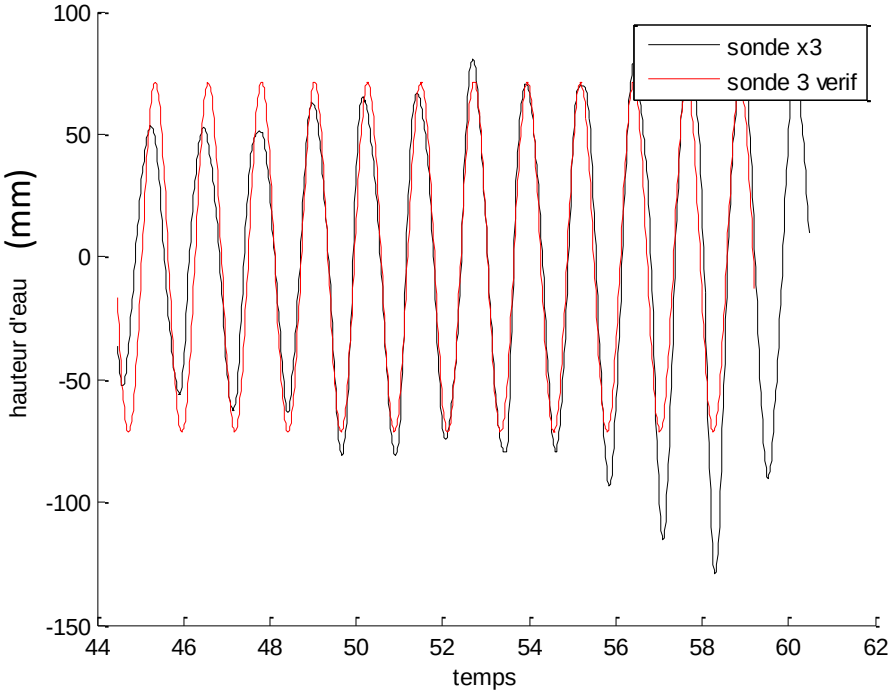
Aval
Signal des sondes



test_3_HS_1_8_Tp_5_5

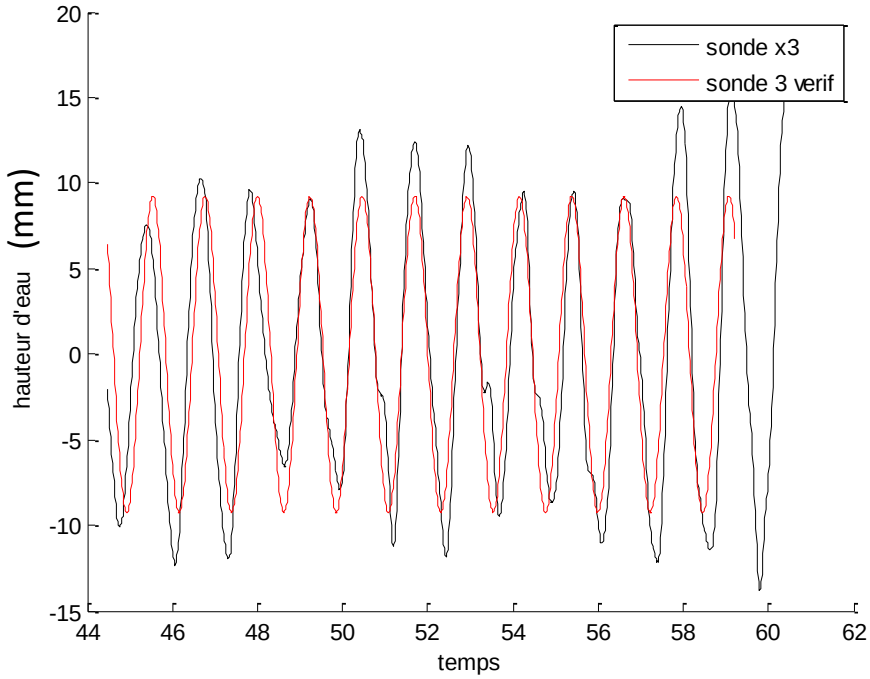
Tm=1.23s

Amont



aincident=54.5885
refl_s3sondes =0.3379

Aval

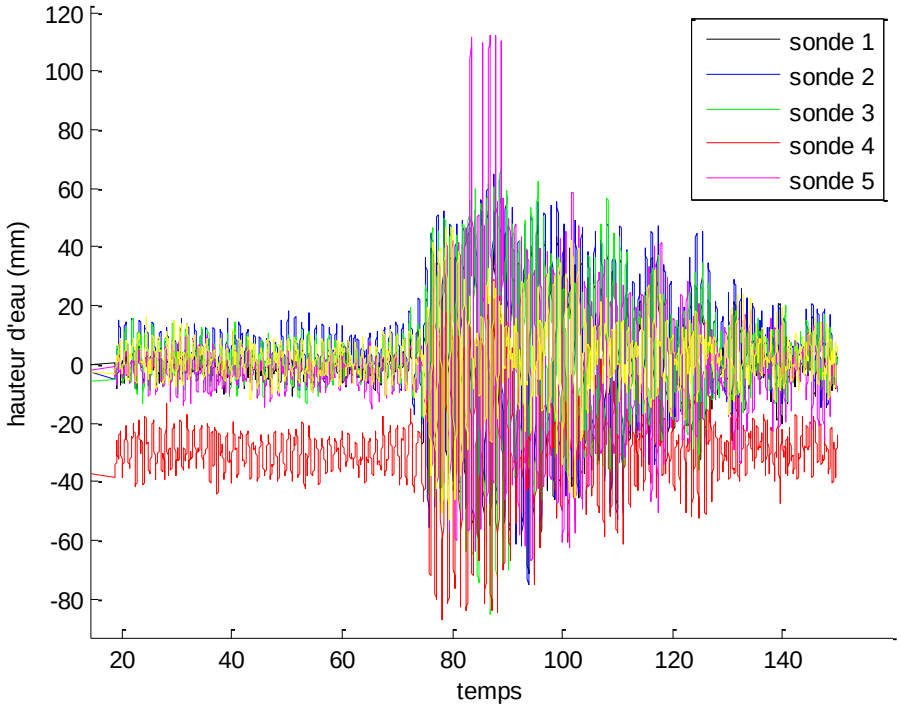


a_transmis=8.0253
refl_s3sondes_plage =0.1873

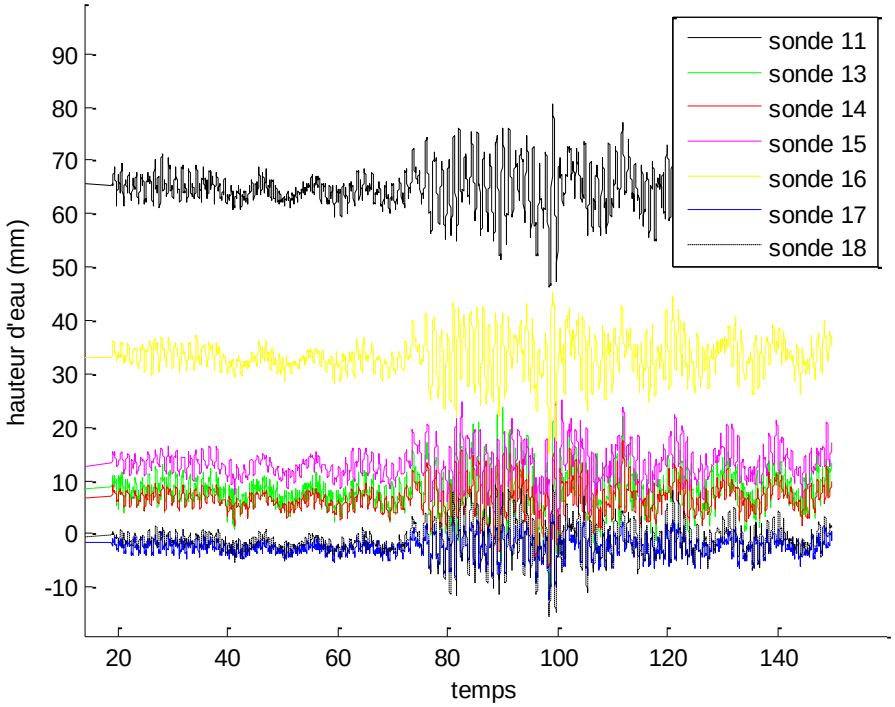
test_4_HS_1_7_Tp_5_0

Tm=1.118s

Amont
Signal des sondes



Aval
Signal des sondes

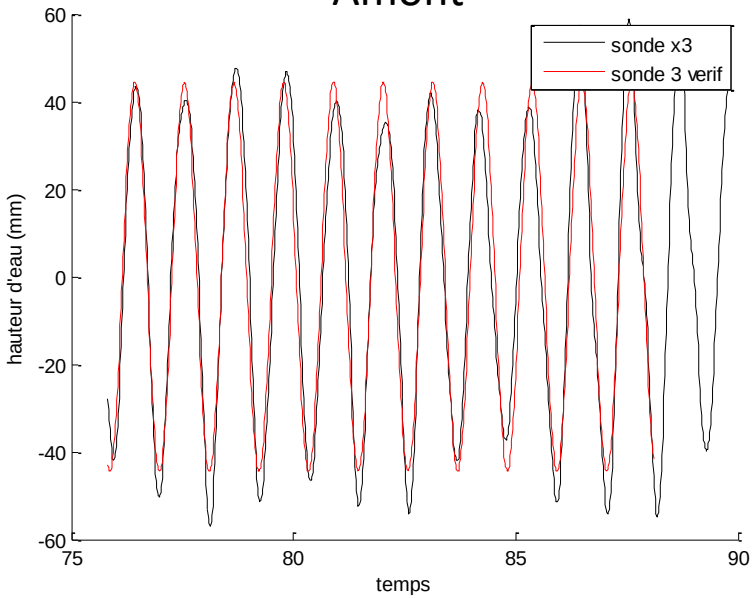


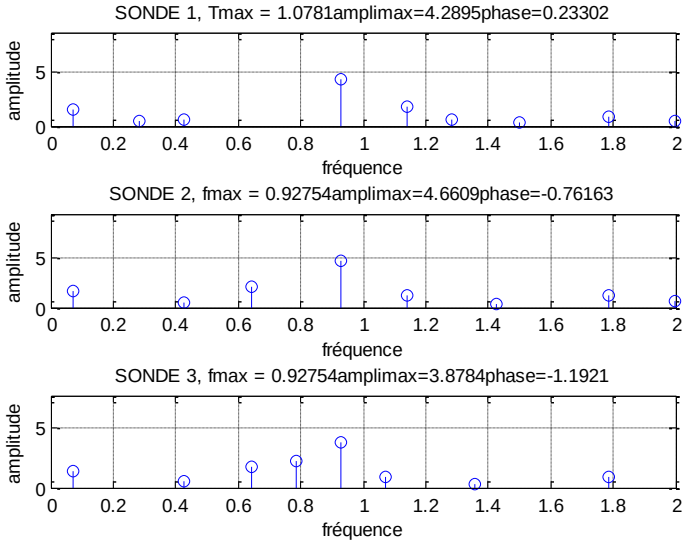
test_4_HS_1_7_Tp_5_0

Tm=1.118s

Aval

Amont



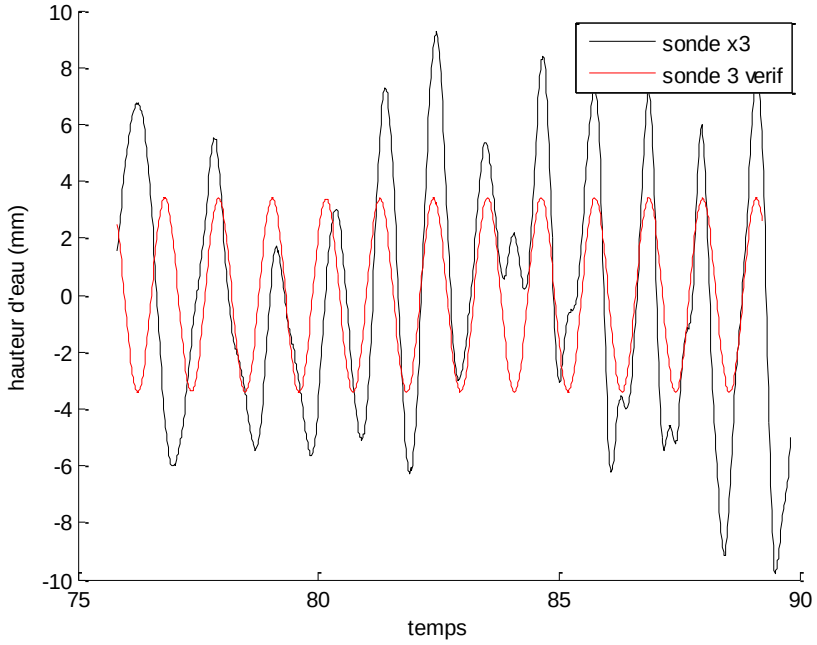


aincident =42.4079
refl_s3sondes =0.2925

Signal « propre » en amont de la structure

atransmis = 4.4089mm
refl_s3sondes_plage = 0.1931mm

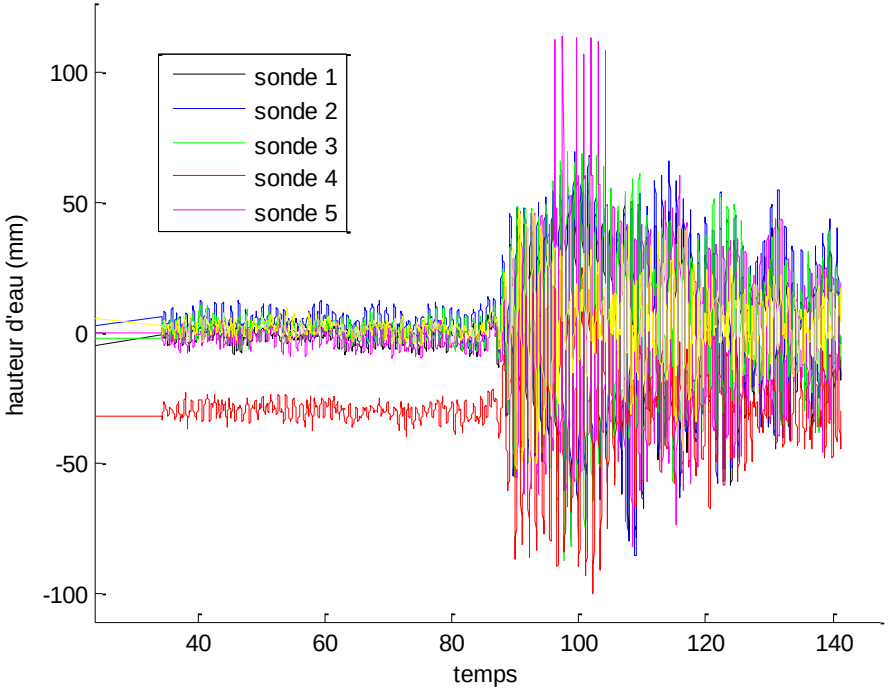
Faible transmission mais seiche en aval qui perturbe l'analyse à Tm.



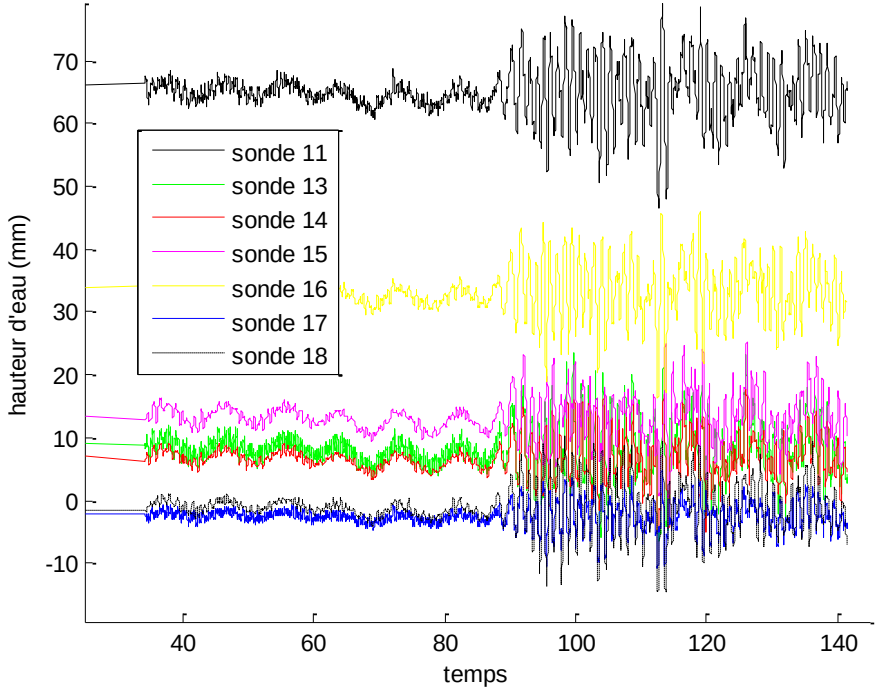
test_5_HS_0_09_Tp_5_0

Tm=1.118s

Amont
Signal des sondes



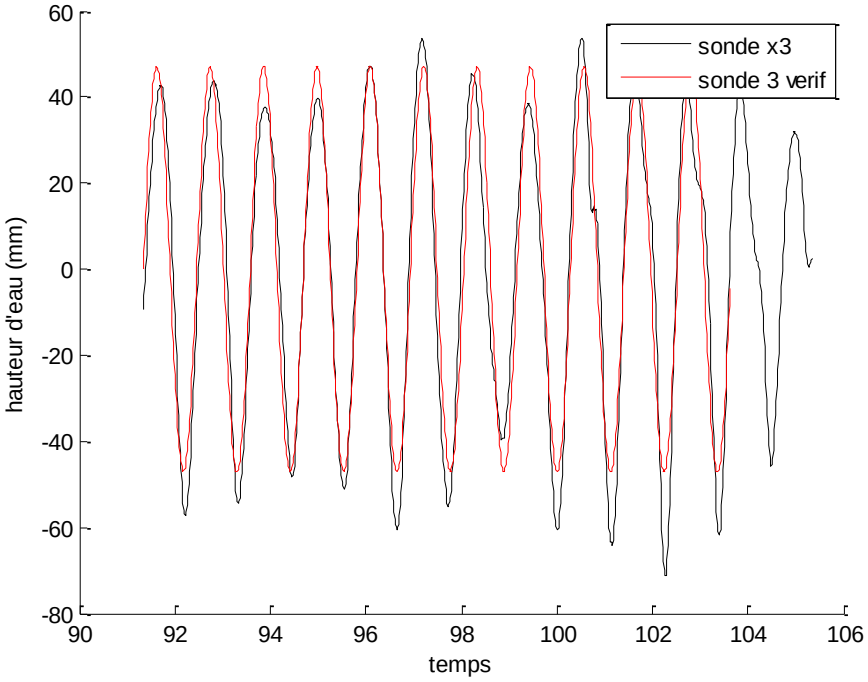
Aval
Signal des sondes



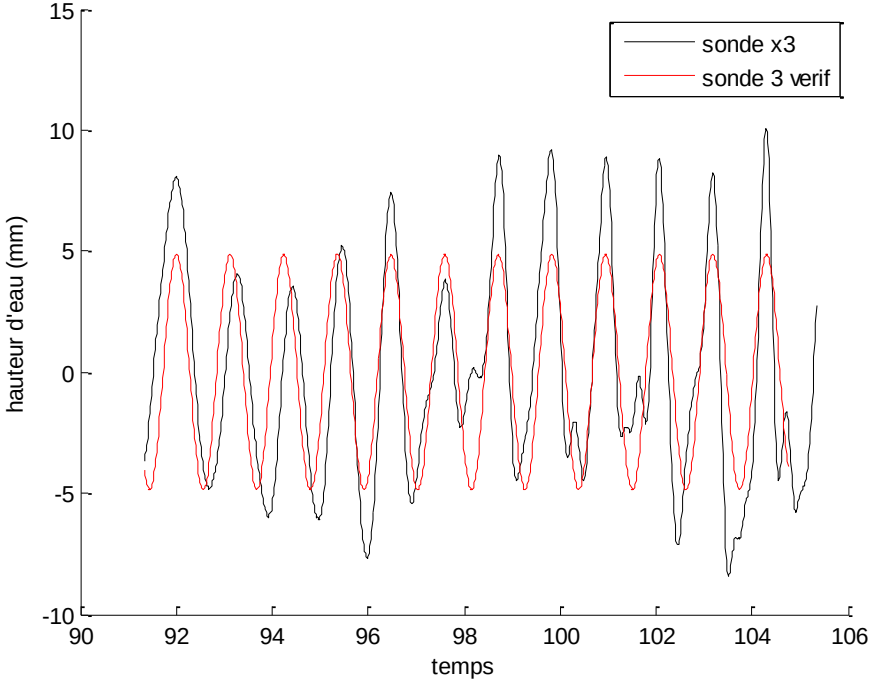
test_5_HS_0_09_Tp_5_0

Tm=1.118s

Amont



Aval



aincident =44.1546mm

a_transmis=6.1096mm

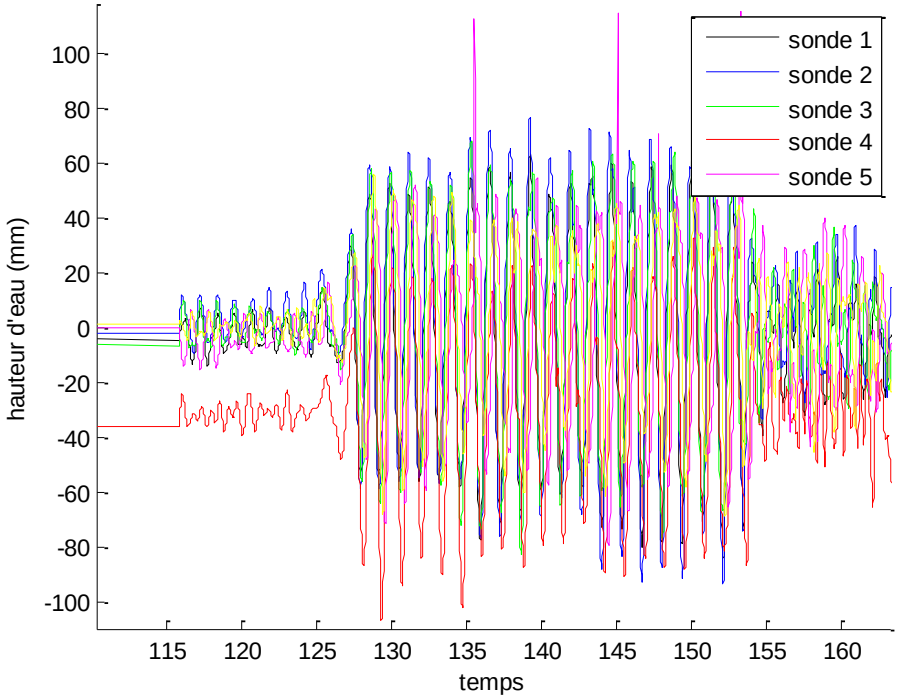
refl_s3sondes = 0.4066

refl_s3sondes_plage =0.2419

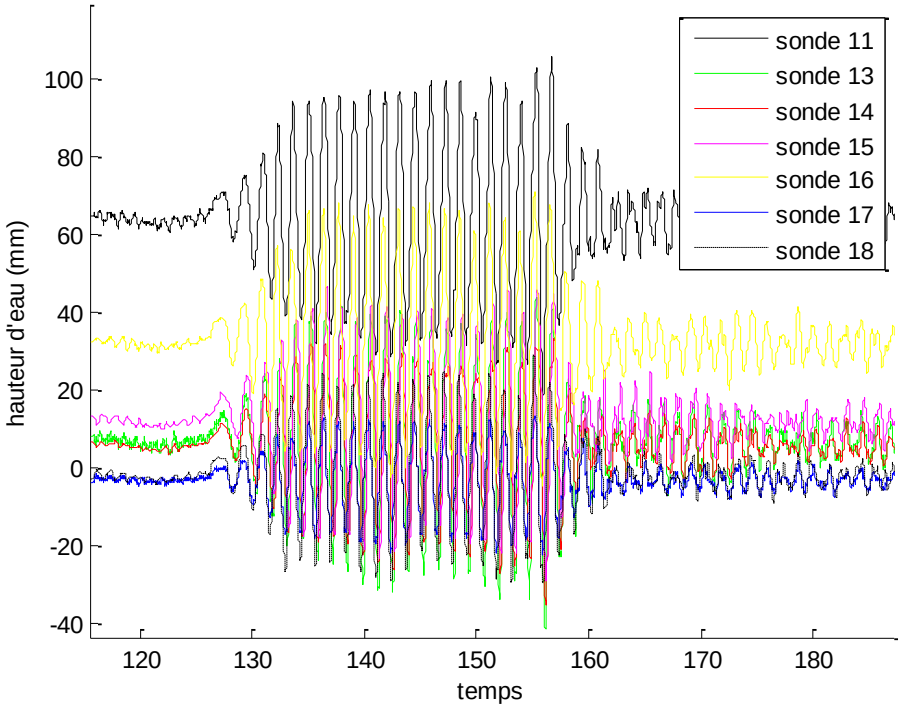
test_6_HS_0_105_Tp_6_0

Tm=1.3416s

Amont
Signal des sondes



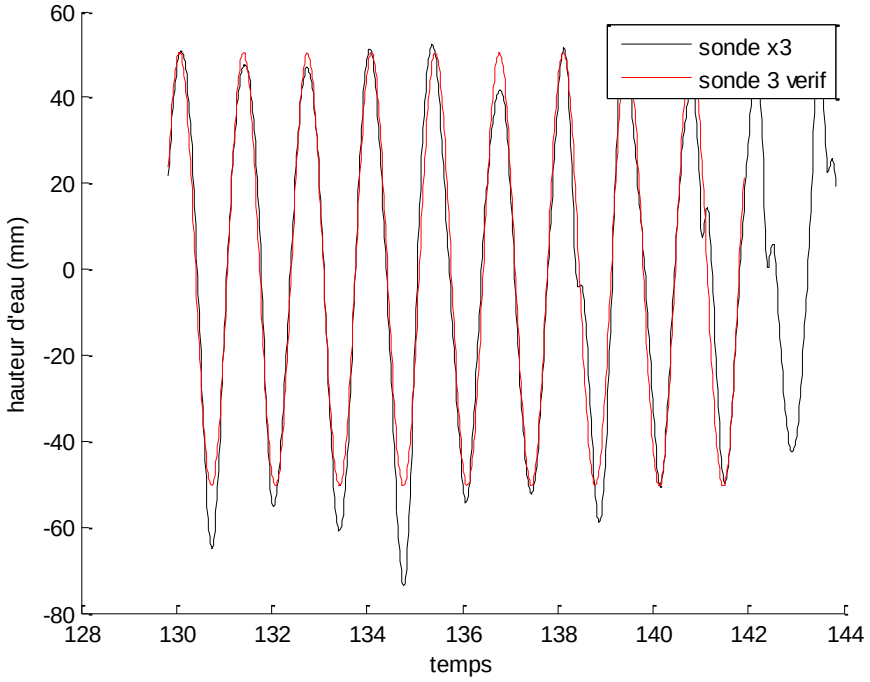
Aval
Signal des sondes



test_6_HS_0_105_Tp_6_0

Tm=1.3416s

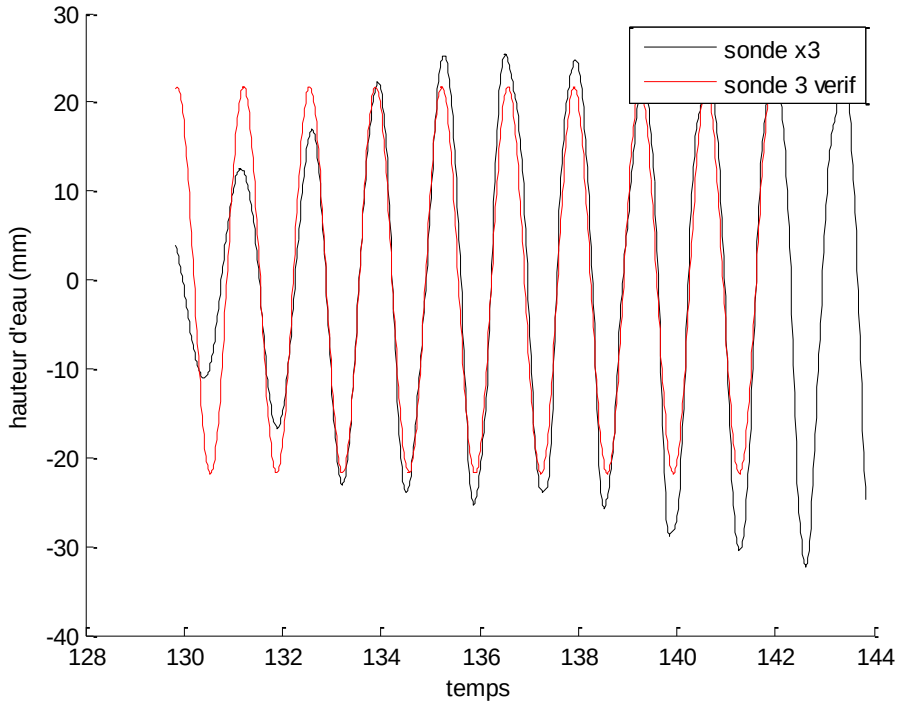
Amont



aincident =53.4036mm

refl_s3sondes =0.2000

Aval



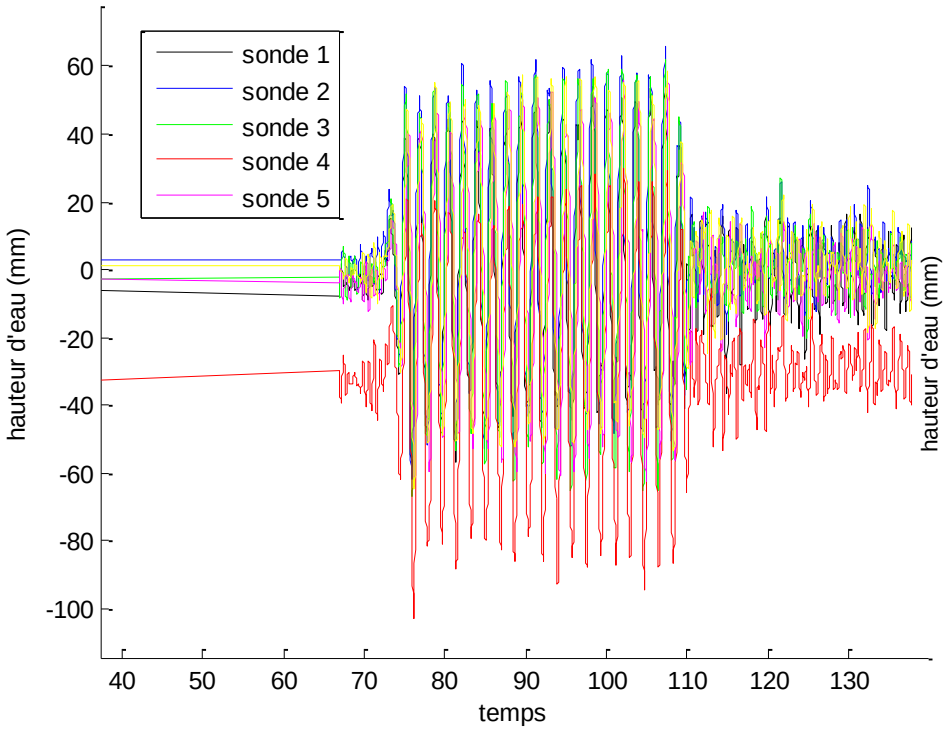
atransmis=24.2436mm

refl_s3sondes_plage =0.1436

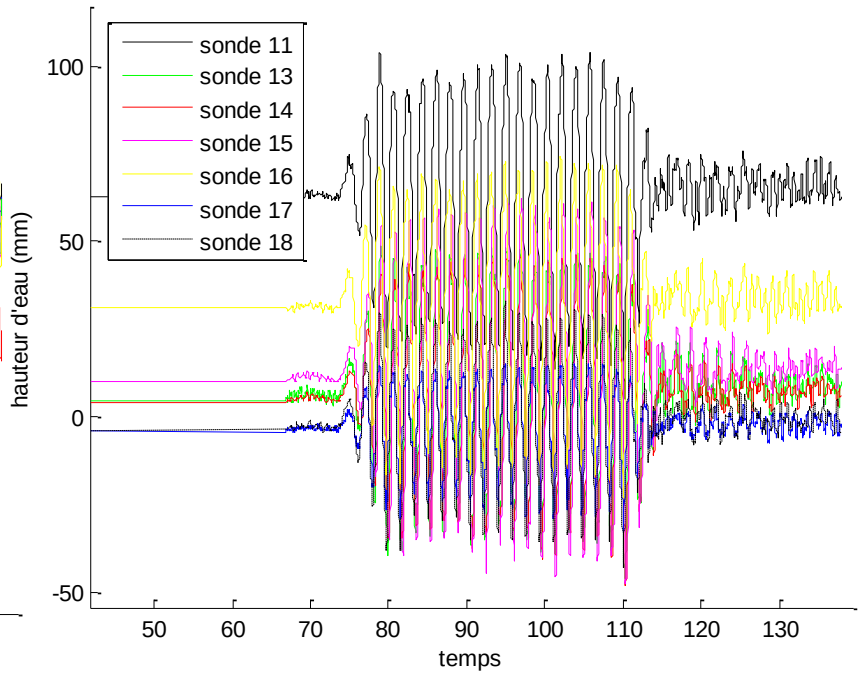
test_7_HS_0_10_Tp_8_0

Tm=1.7889s

Amont
Signal des sondes



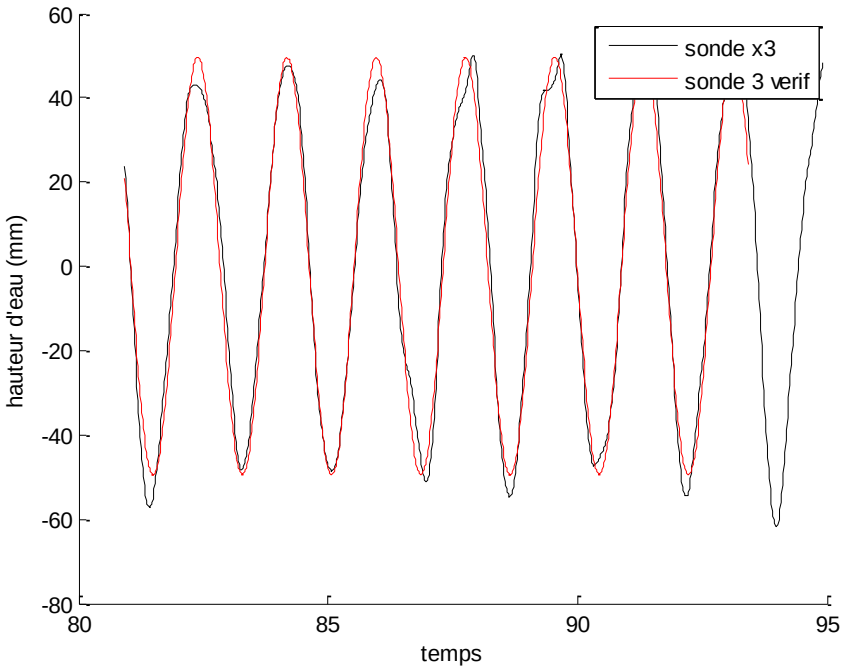
Aval
Signal des sondes



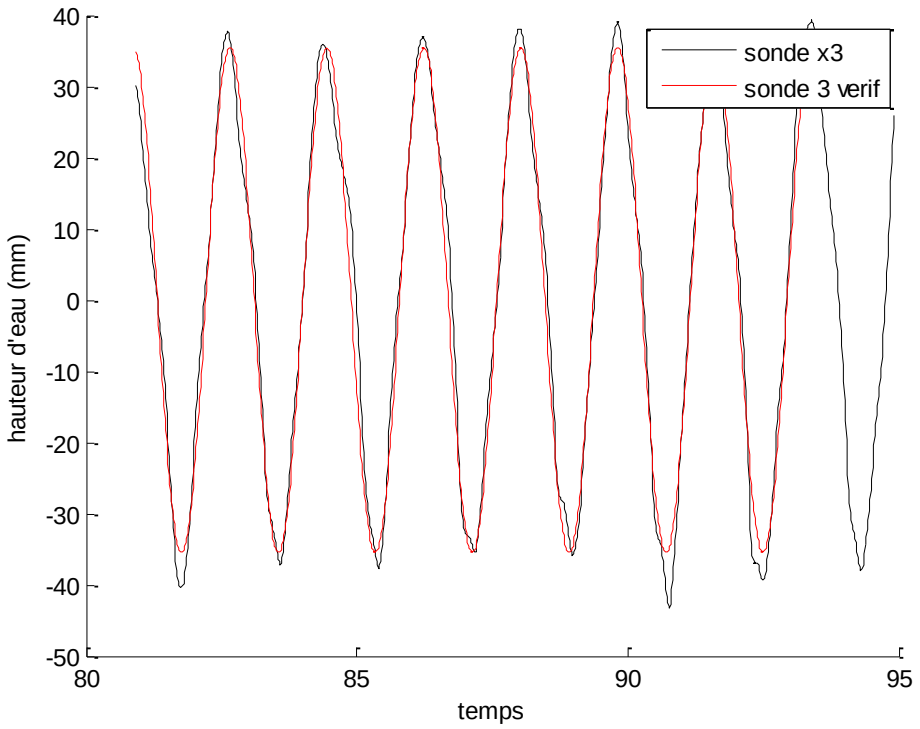
test_7_HS_0_10_Tp_8_0

Tm=1.7889s

Amont



Aval



aincident=51.0846mm
refl_s3sondes =0.0447

Faible réflexion

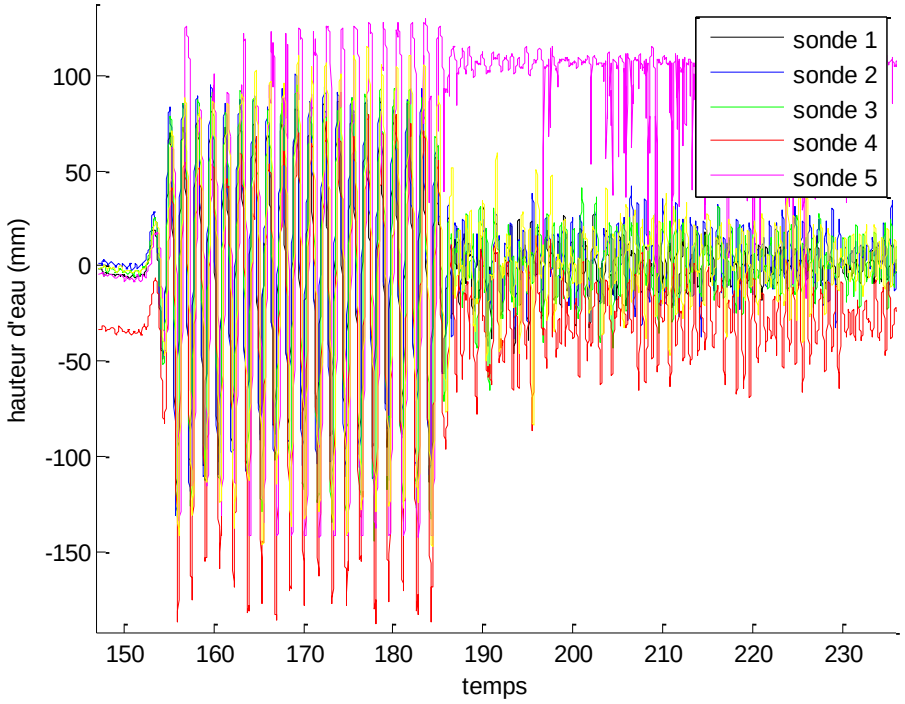
a_transmis=37.2400mm

refl_s3sondes_plage =0.0081

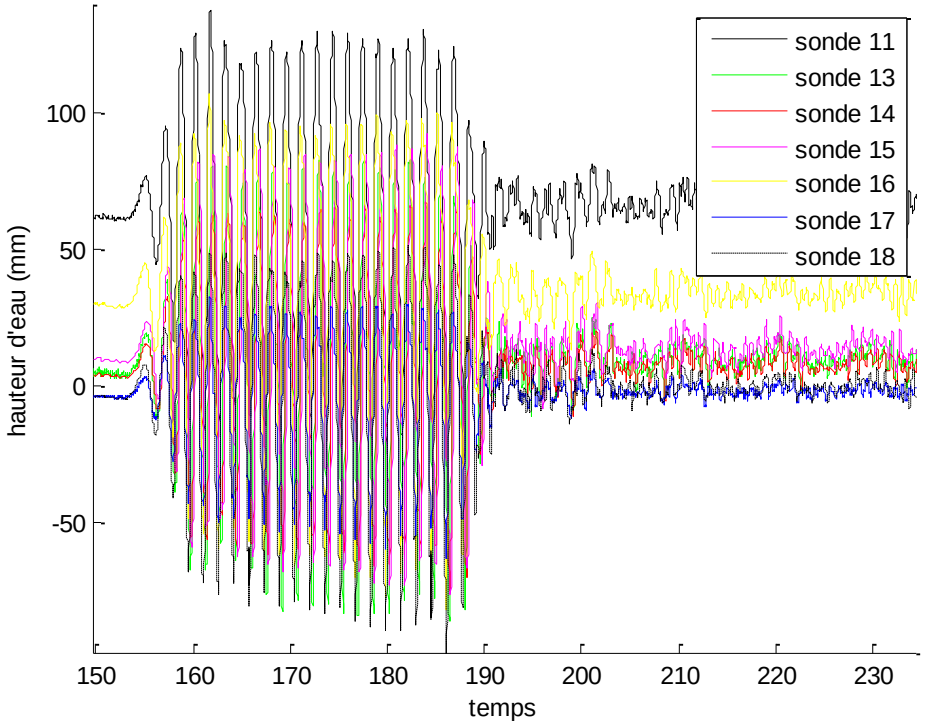
test_8_HS_0_0985_Tp_7_0

Tm= 1.5652s

Amont
Signal des sondes



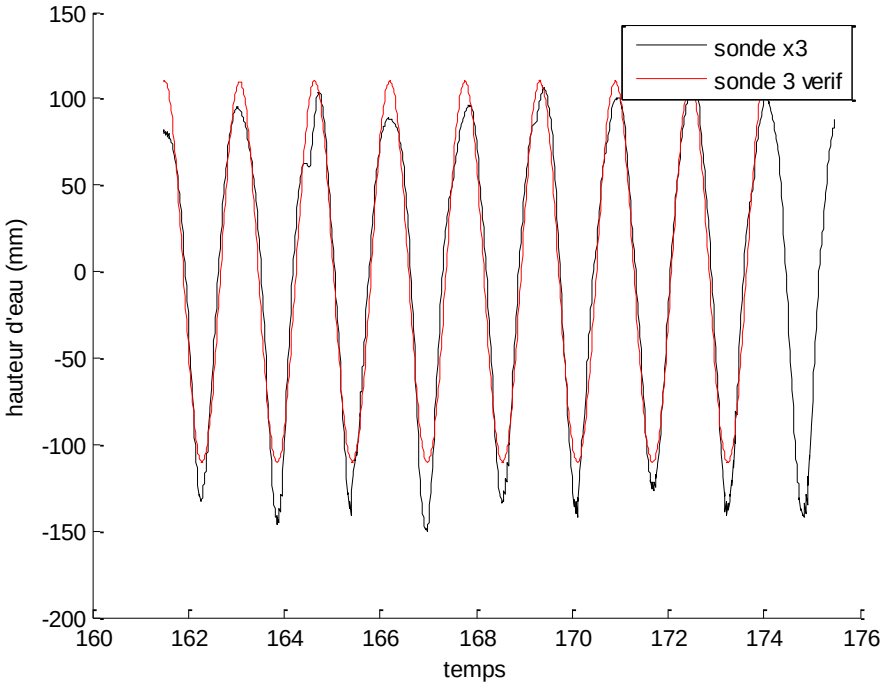
Aval
Signal des sondes



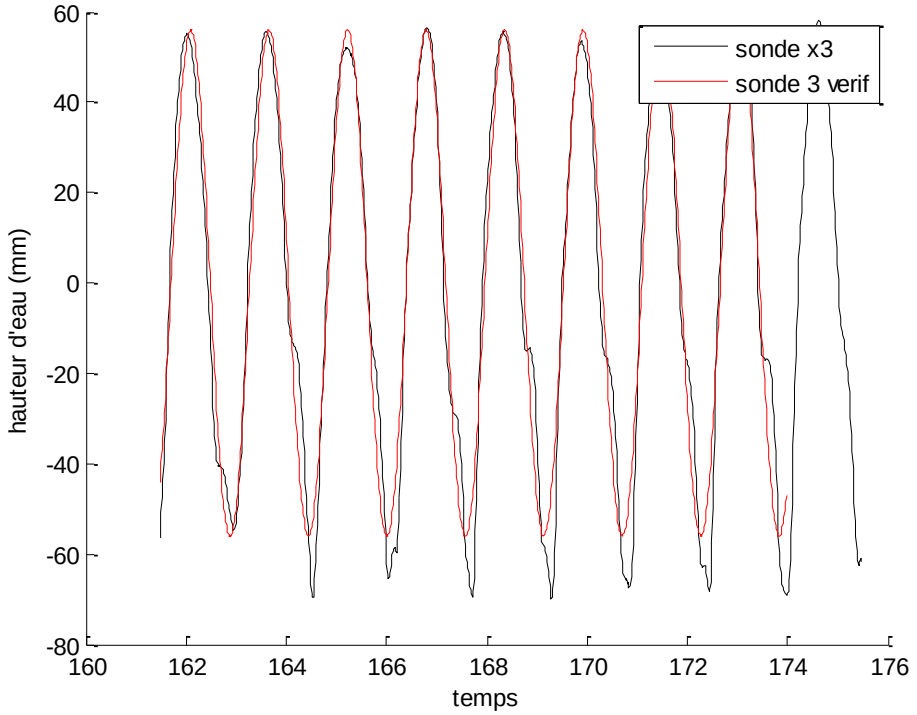
test_8_HS_0_0985_Tp_7_0

Tm= 1.5652s

Amont



Aval



aincident =98.7263mm

a_transmis= 65.6808mm

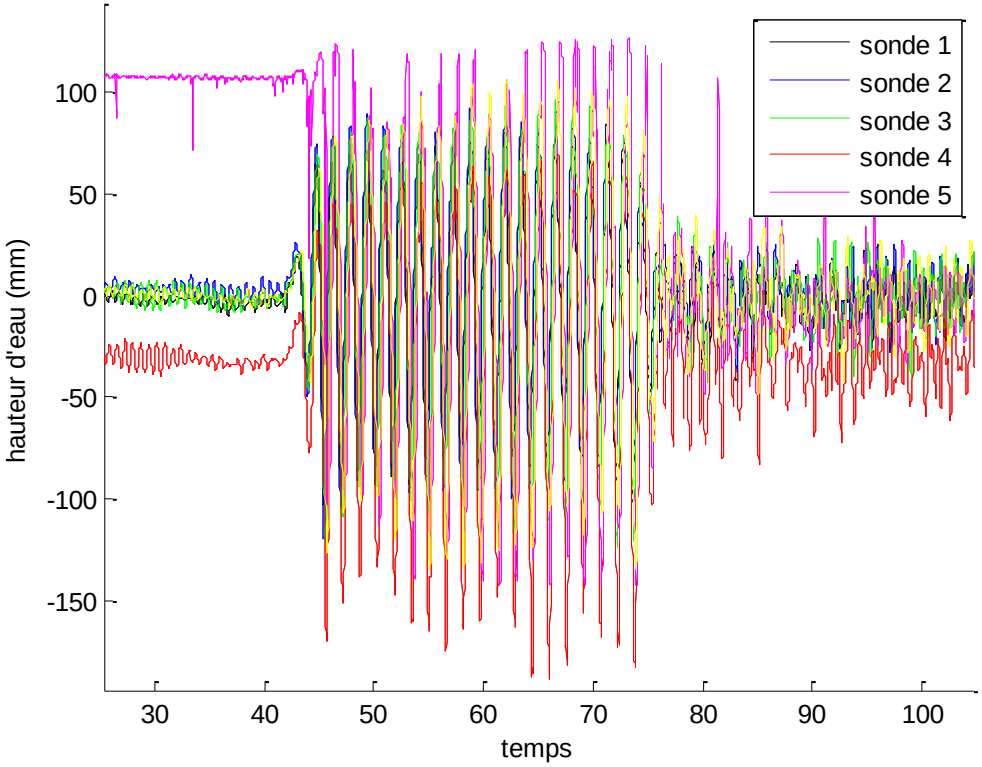
refl_s3sondes = 0.1511

refl_s3sondes_plage =0.1063

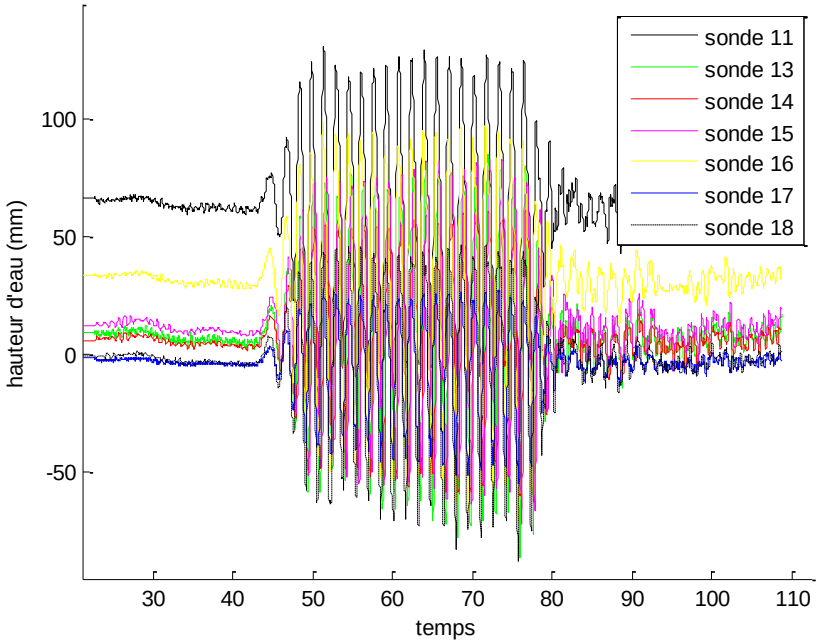
test_9_HS_0_0875_Tp_7_0

Tm= 1.5652s

Amont
Signal des sondes



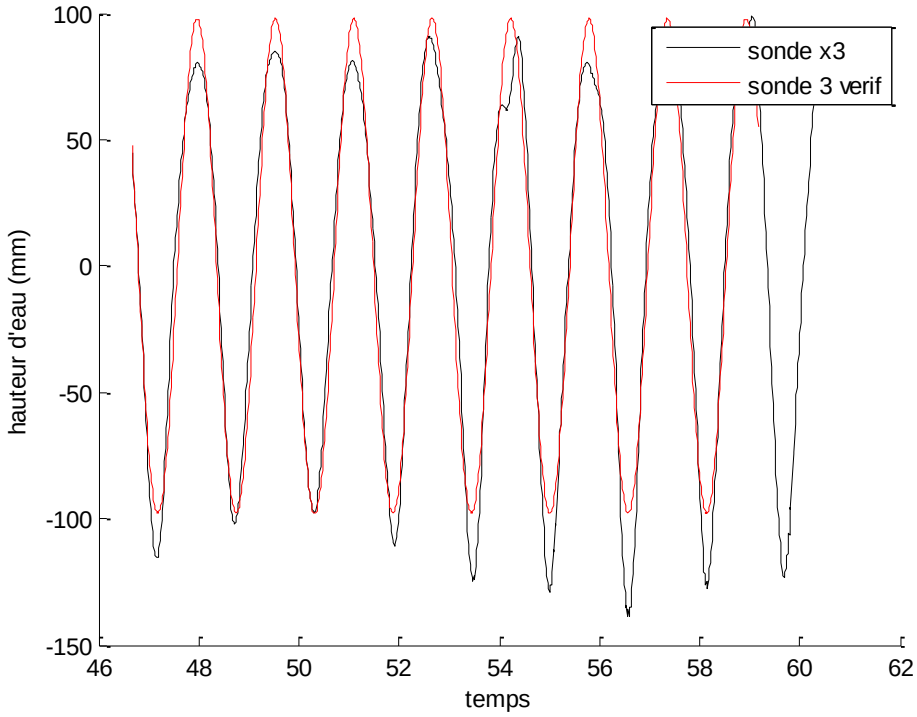
Aval
Signal des sondes



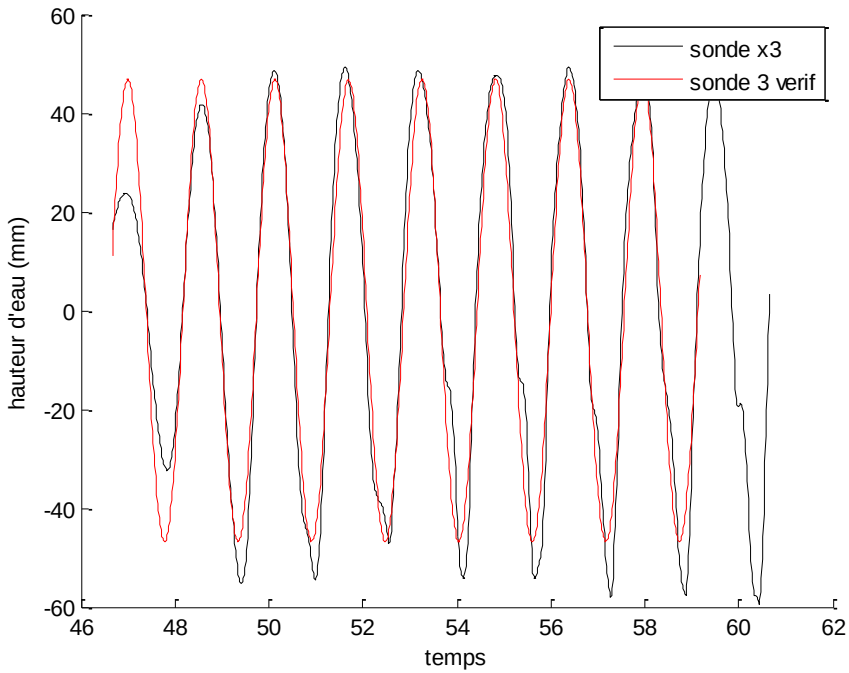
test_9_HS_0_0875_Tp_7_0

Tm= 1.5652s

Amont



Aval



aincident=88.4706mm

refl_s3sondes =0.1237

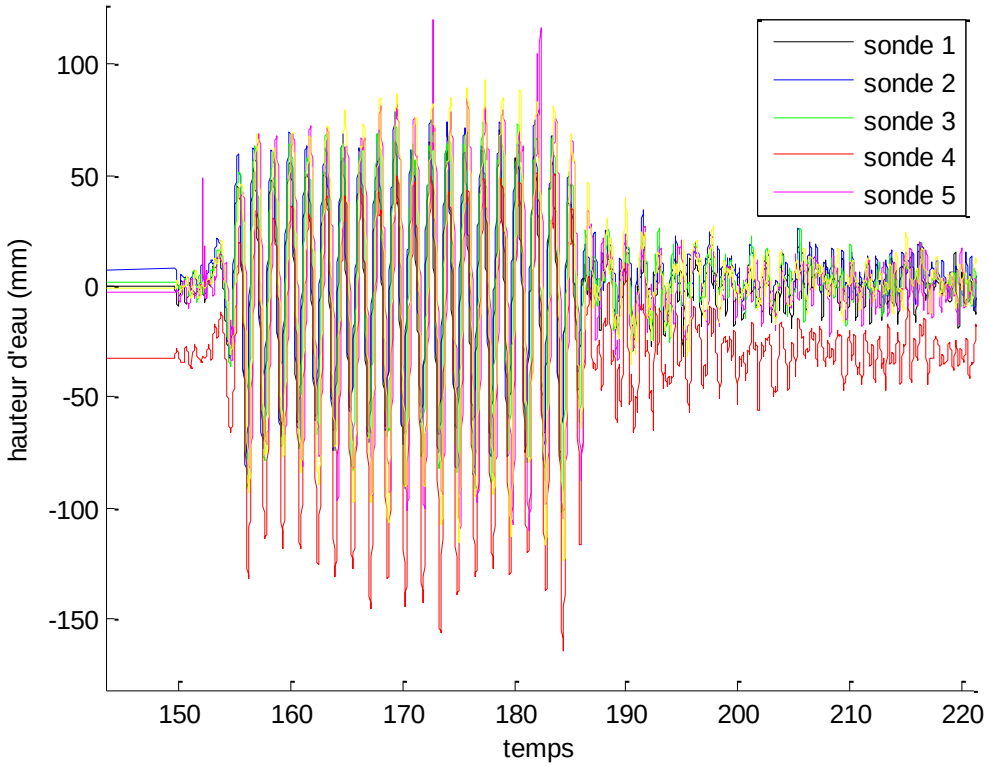
atransmis=55.5353

refl_s3sondes_plage =0.1226

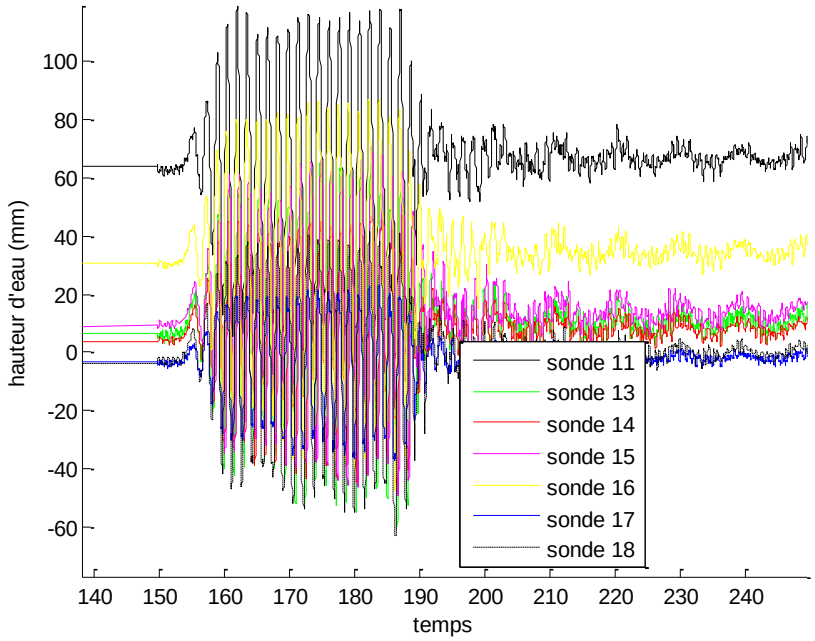
test_10_HS_0_0675_Tp_7_0

Tm= 1.5652s

Amont
Signal des sondes



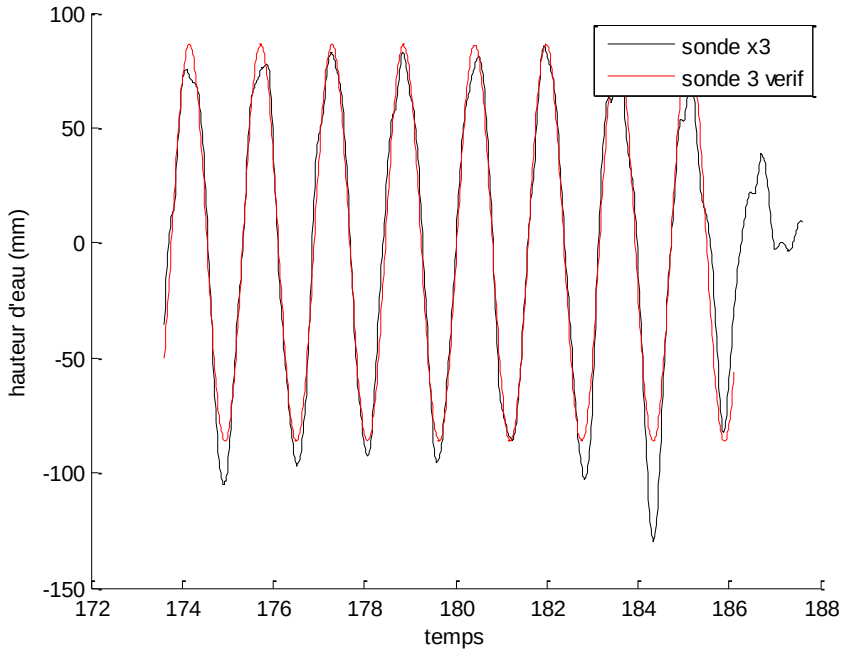
Aval
Signal des sondes



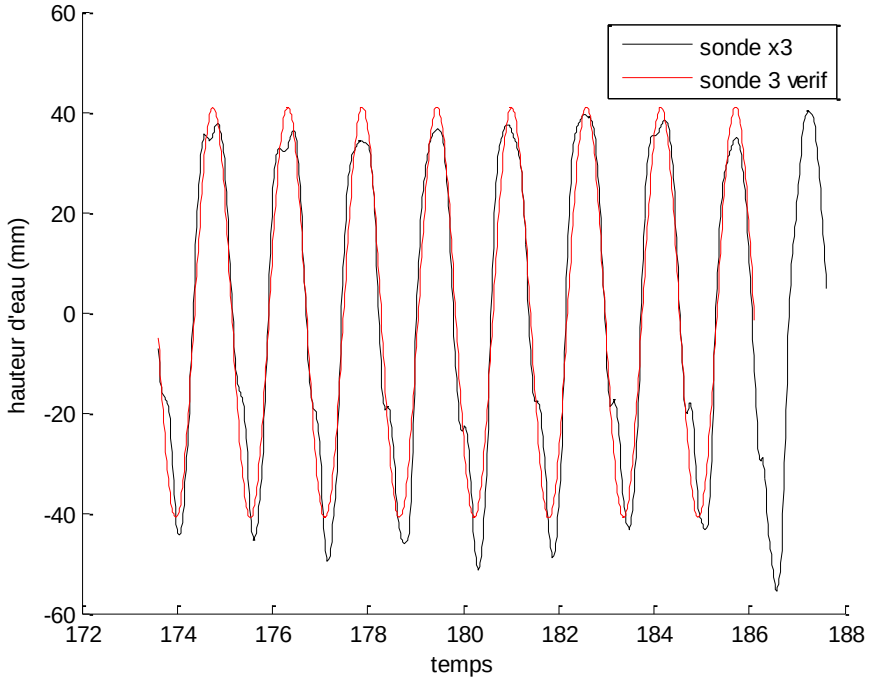
test_10_HS_0_0675_Tp_7_0

Tm= 1.7889s

Amont



Aval



aincident =74.2802 mm

atransmis=50.1478mm

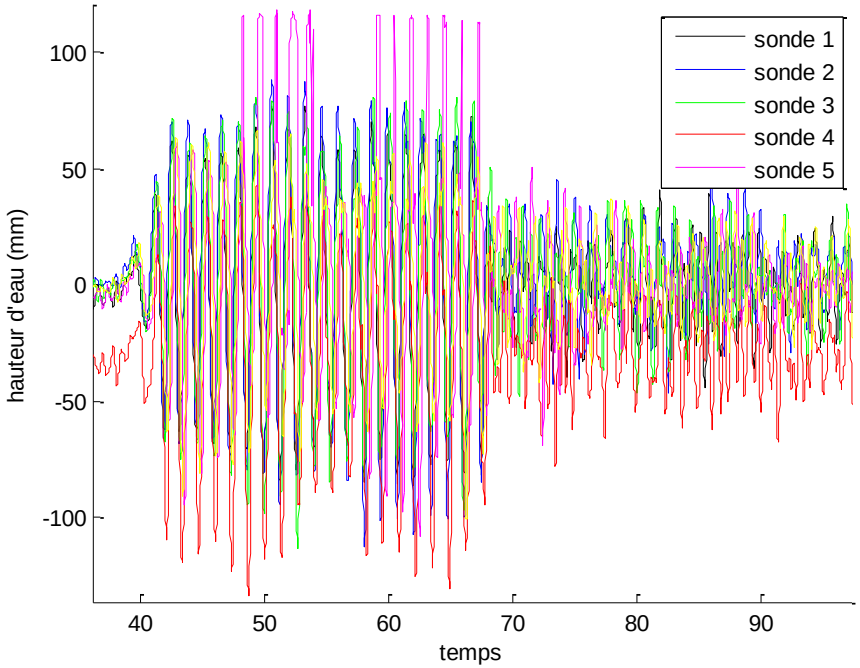
refl_s3sondes =0.2117

refl_s3sondes_aval =0.1428

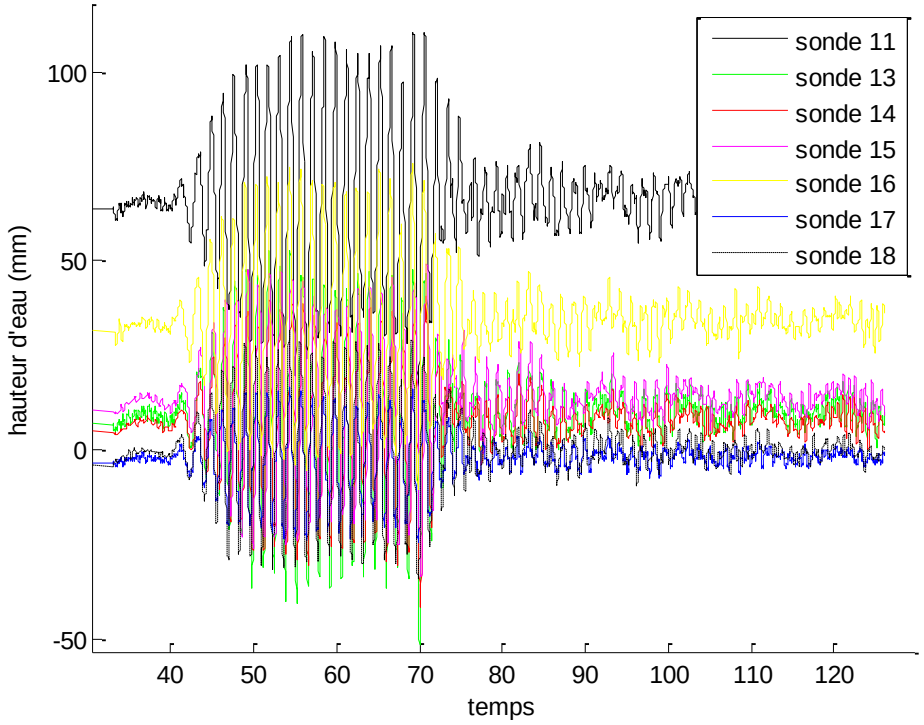
test_11_HS_0_0650_Tp_6_0

Tm= 1.3416s

Amont
Signal des sondes



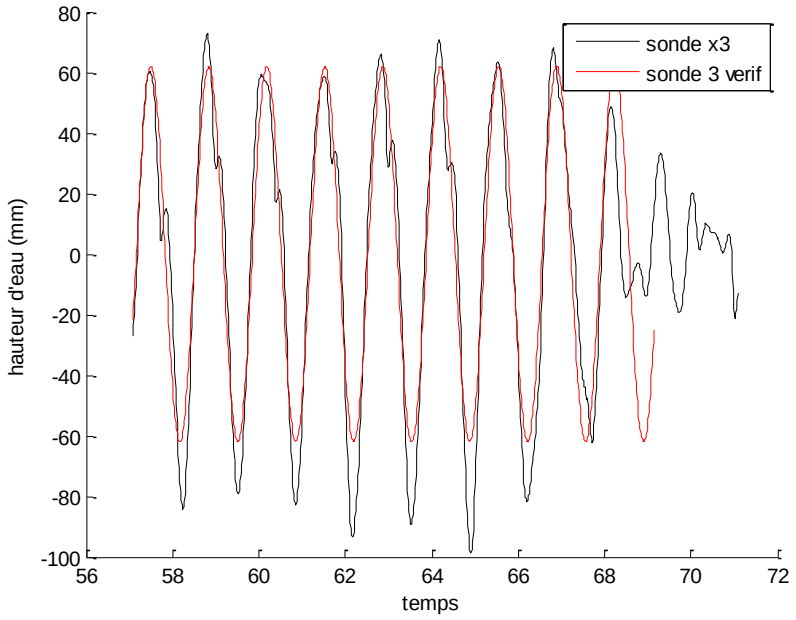
Aval
Signal des sondes



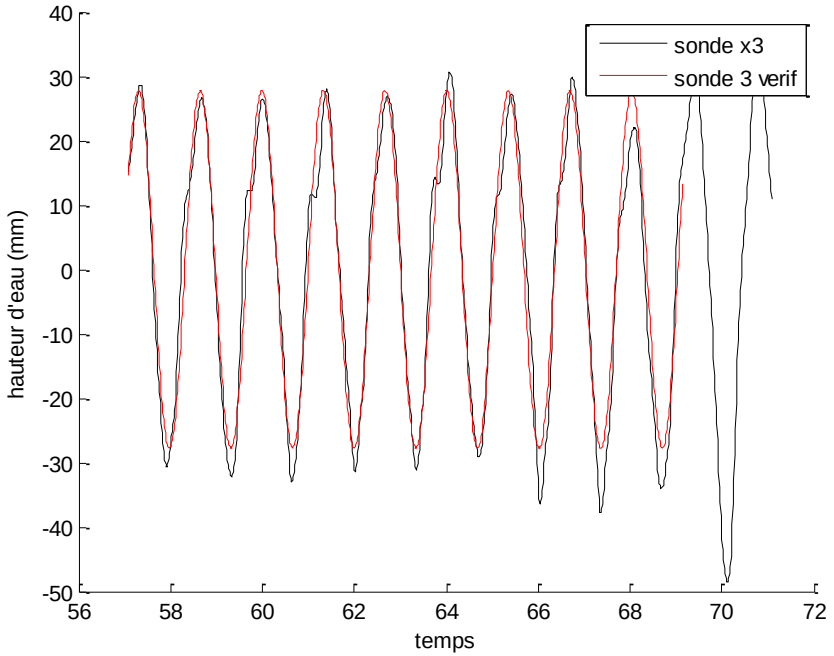
test_11_HS_0_0650_Tp_6_0

Tm= 1.3416s

Amont



Aval



aincident =62.8635 mm

atransmis = 30.4191mm

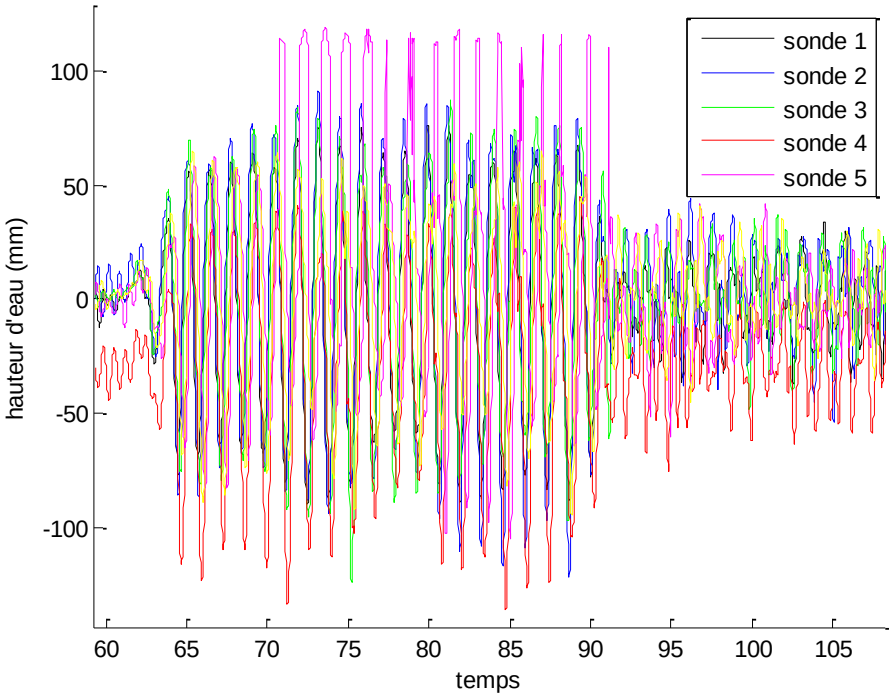
refl_s3sondes =0.2042

refl_s3sondes_plage=0.2035

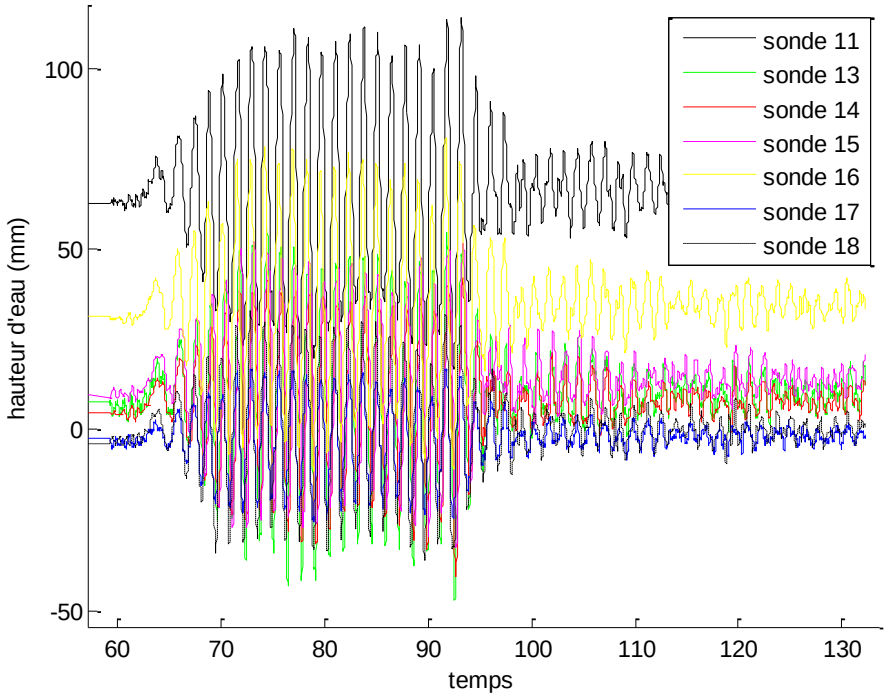
test_12_HS_0_0673_Tp_6_0

Tm= 1.3416s

Amont
Signal des sondes



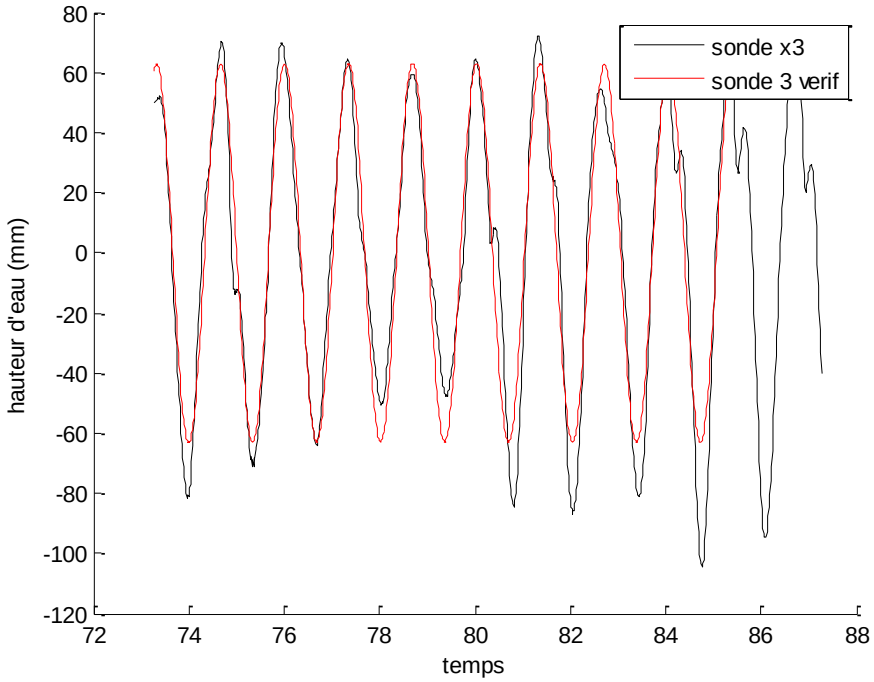
Aval
Signal des sondes



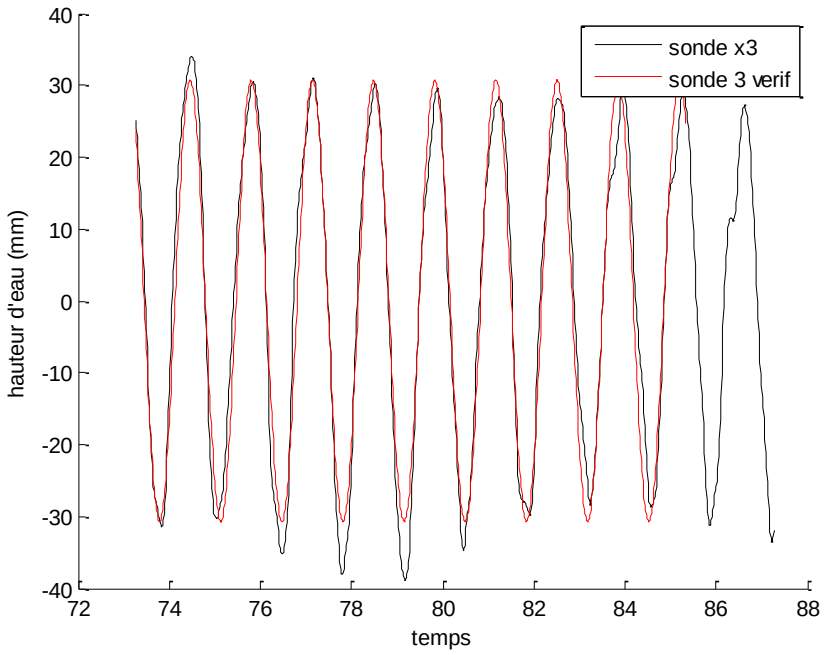
test_12_HS_0_0673_Tp_6_0

Tm= 1.3416s

Amont



Aval



aincident=67.0270 mm

refl_s3sondes = 0.2249

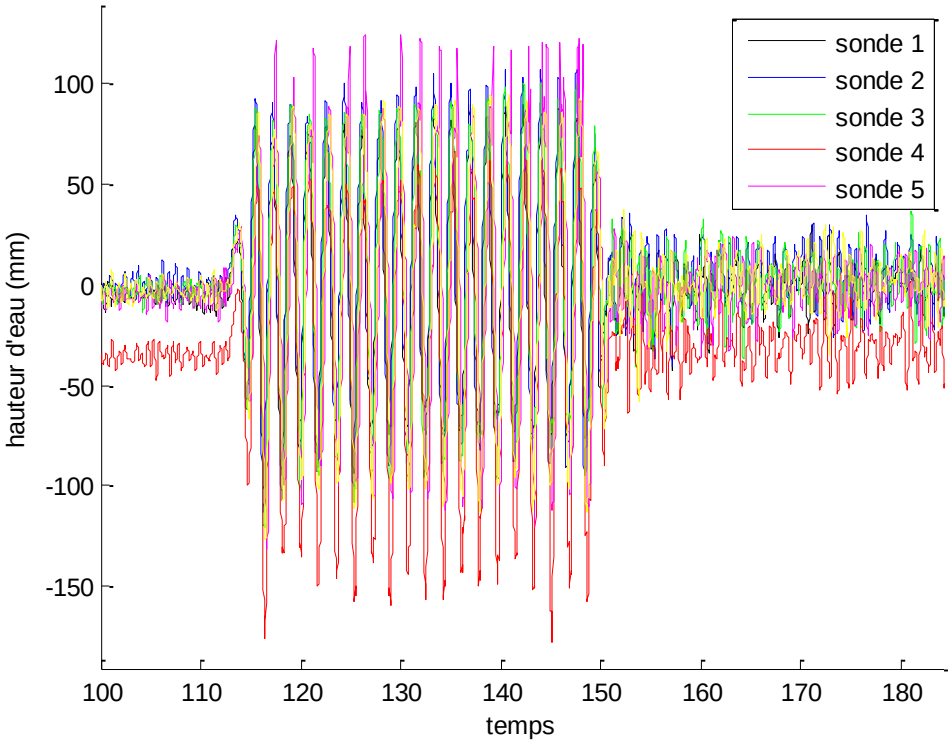
atransmis = 33.3919mm

refl_s3sondes_plage =0.2254

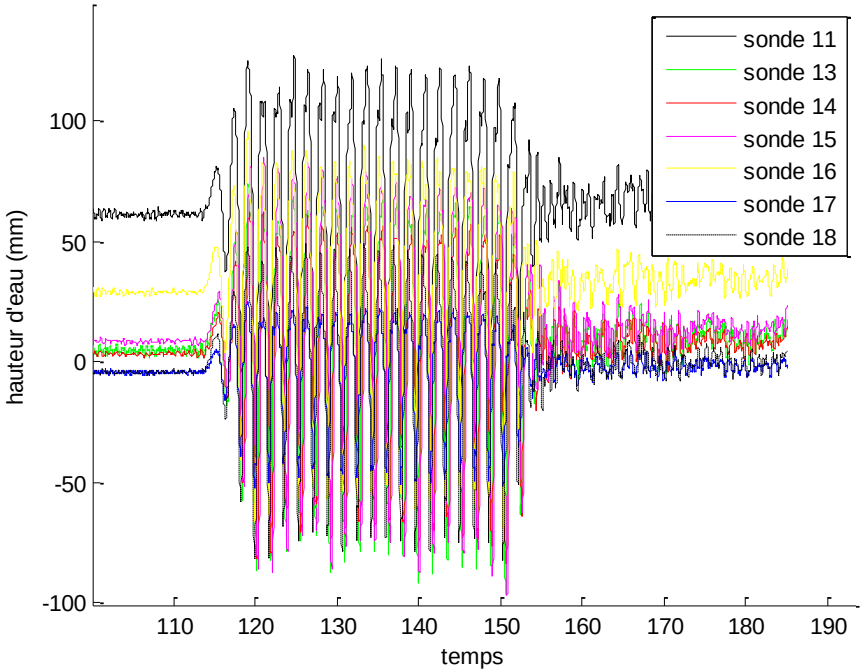
test_13_HS_0_0925_Tp_8_0

Tm= 1.7889s

Amont
Signal des sondes

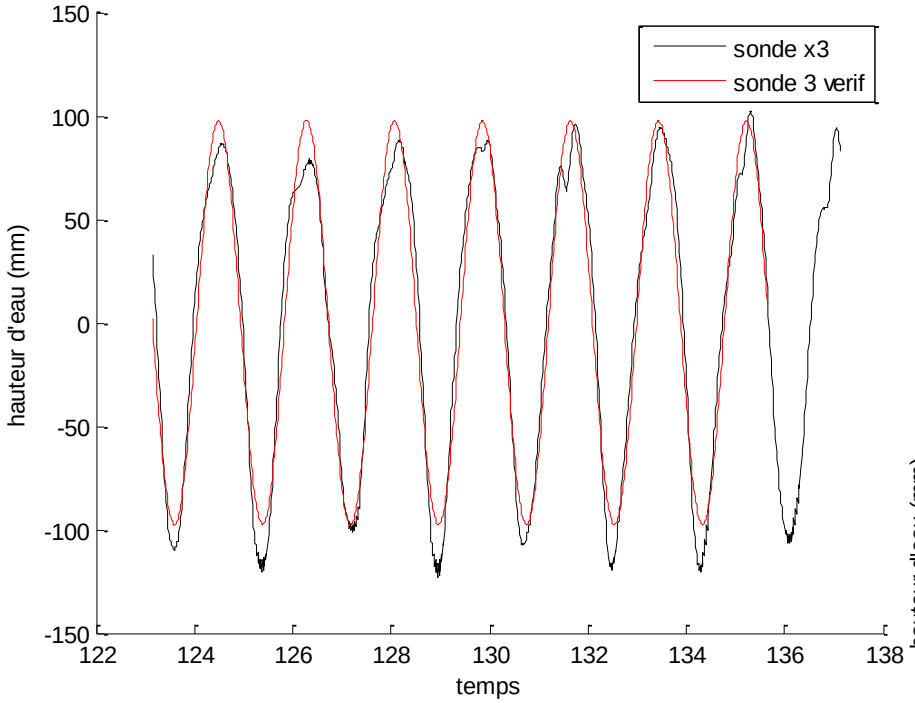


Aval
Signal des sondes



test_13_HS_0_0925_Tp_8_0

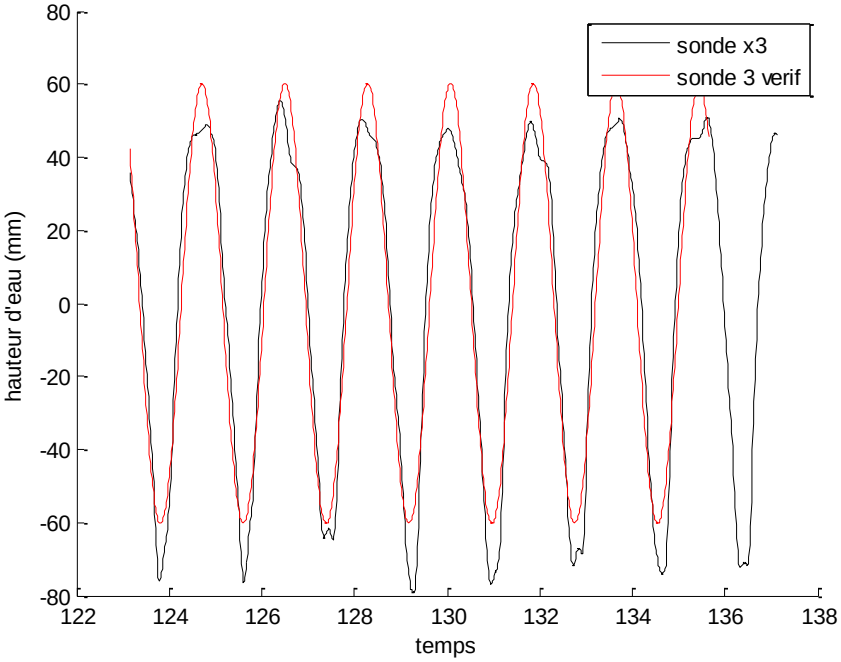
Tm= 1.7889s



aincident =
92.2386

refl_s3sondes =0.0835

Aval

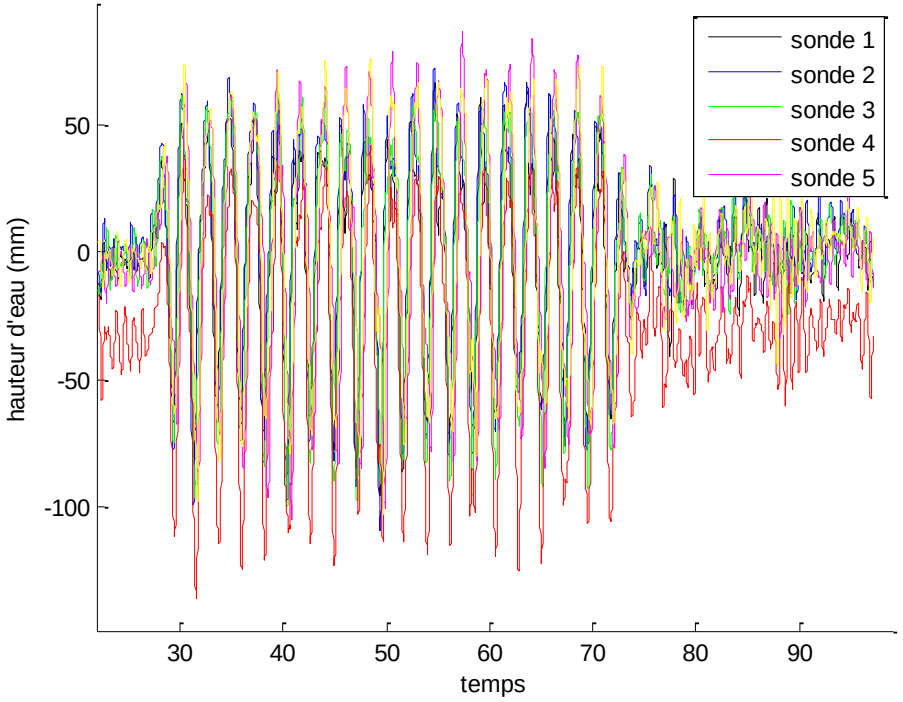


atransmis=61.4733mm
refl_s3sondes_aval =0.0469

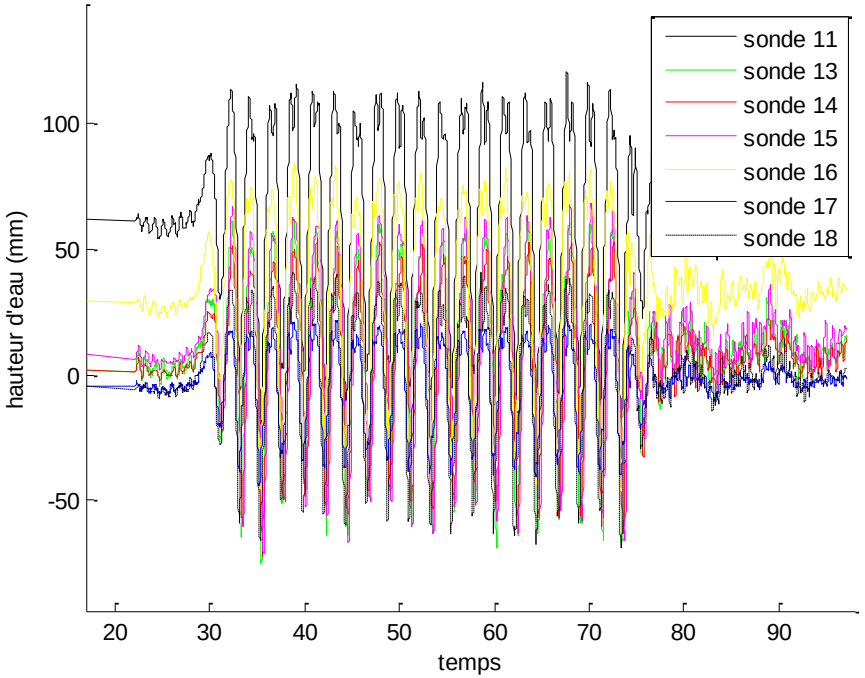
test_14_HS_0_0875_Tp_10_0

Tm= 2.2361s

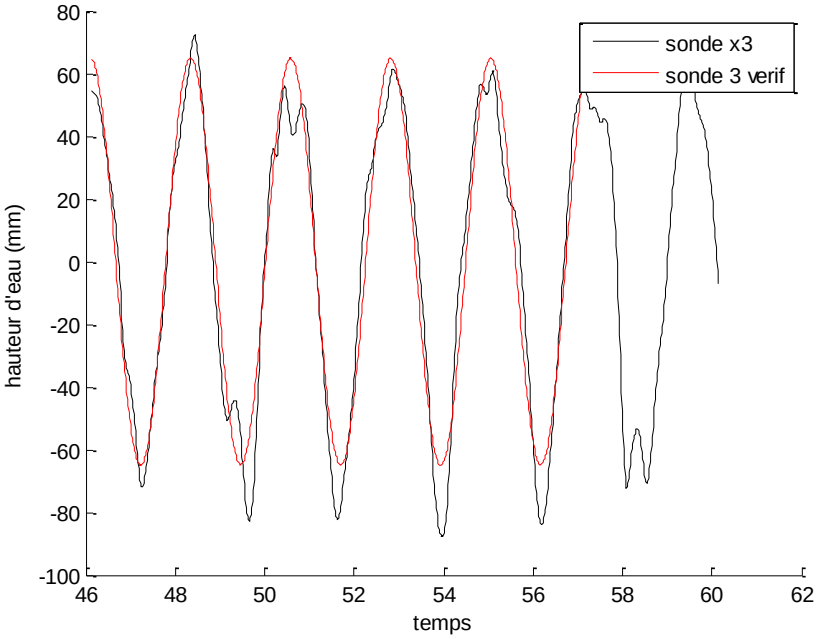
Amont
Signal des sondes



Aval
Signal des sondes



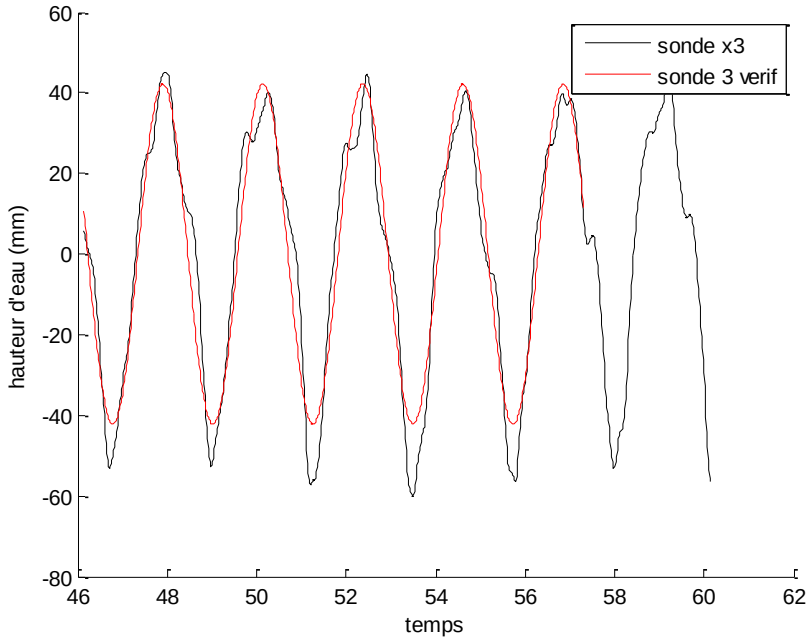
Amont



aincident =61.6787

refl_s3sondes = 0.1109

Aval



atransmis =47.7687

refl_s3sondes_plage =0.2024